Autonomous Mobile Robot

X300B Serial User Manual V1.1

Intelligent. Efficient. Simple

Version Record

Revised Date	Version	Revised Content	Editor	Auditor
20250224	V1.0.0	X300B-L First version released	Peng Guo	Chenxi Jiang
20250514	V1.1.0	Added X300B, X300T	Weilin Wang	Chenxi Jiang
20251103	V1.3.0	Add description of functions such as optimizing multi-floor maps, task configuration, and Settings	Weilin Wang	Chenxi Jiang

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Introduction

Thank you for choosing our mobile robots. These robots integrate advanced autonomous navigation technology, intelligent perception technology and efficient electromechanical components, and are designed to provide intelligent, efficient, simple, safe and reliable intelligent handling solutions for industrial, service and other scenarios. In order to help you quickly learn the robot functions and use them safely and efficiently, please read the user manual carefully before operation.

Applicable objects:

All robot users and people who interact with robot in the scene.

Product design concept: 1-2-N-X

iBEN mobile robots are developed based on a set of general software and hardware architectures and modular design concepts. Through multi-sensor fusion (such as lidar, depth camera, visual camera, IMU, etc.), AI algorithm optimization, large model application and cloud collaborative management, the robot can achieve strong perception mapping navigation without deployment, precise positioning, dynamic obstacle avoidance, path planning and multi-robot collaboration to meet the diverse needs in complex environments.

iBEN intelligent mobile robots have developed two major business methods for different types of customers and scenarios:

1. Human-robot interaction solutions suitable for flexible scenarios of small and medium-sized enterprises: high flexible scenarios, semi-automatic and manual command operations through single-robot intelligence + industrial Internet of Things devices.

2. Fully automatic solutions suitable for medium and large industrial integrated scenarios: providing robot chassis navigation and delivery capabilities, meeting scenario adaptation and scalability: robot API interface, hardware power supply system & communication interface, mechanical reserved positions, etc., used to customized models and functional requirements of industrial automation, helping customers create full-scenario flexible delivery solutions.

At the same time, iBEN ROBOT will gradually develop N mobile robots with different loads to meet the needs of different industries, and will expand X product forms based on standard models (lifting type, lurking type, roller type, ground bracket docking handling, etc.) to assist in intelligent delivery.

Warning safety instructions

Safety is the core principle of product design. During use, please be sure to follow the following principles:

Operating environment inspection: Ensure that there are no unrecognizable obstacles or sources of danger (such as stagnant water, flammable materials) in the robot's working area.

Personnel training: All operators must undergo basic training and be familiar with the use of safety devices such as emergency stop buttons and warning lights.

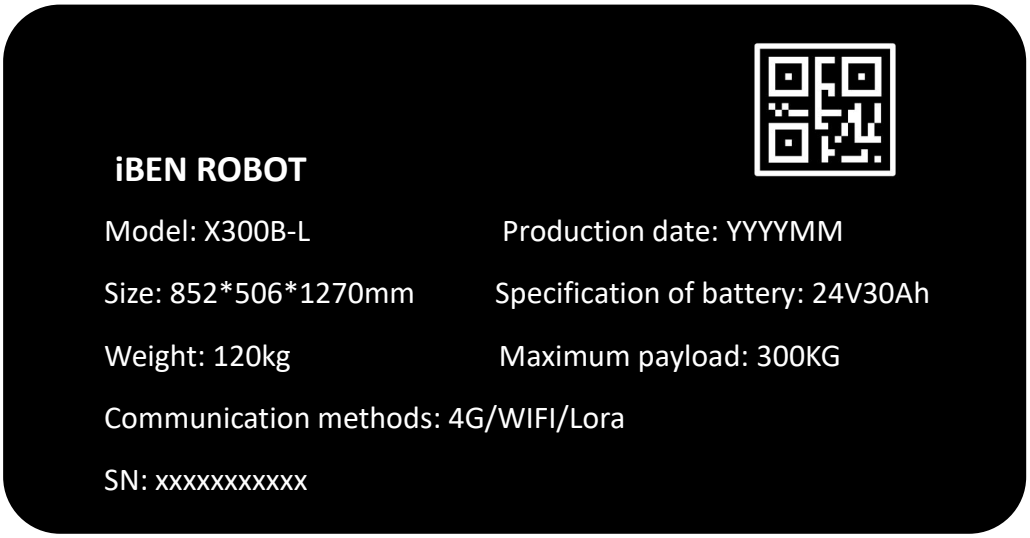
Regular maintenance: Maintain robot according to the manual, replace worn parts timely, and avoid accidents caused by mechanical failures.

The following marks are used in the manual to describe important safety matters. Please be sure to read the instructions carefully.

Warning sign	Related issues
 CAUTION/WARNING	● Severe personal injury or death or unsafe operation may result.
	● After starting up, the robot is dispatched by the control system and may enter the working mode from the standby mode at any time. Therefore, please keep a safe distance from the robot at all times and try to avoid entering the operating area.

	● The robot have moving or rotating parts. When maintaining or repairing the robot's internal parts, please disconnect the lithium battery first to avoid the risk of scratching or squeezing your fingers.
	● The robot is powered by an internal battery. When maintaining or repairing the robot's internal components, the lithium battery should be disconnected first to avoid the risk of electric shock.

Label



1.Safety instruction

1.1 Electricity instruction

- It is strictly forbidden to charge the robot with non-original chargers. If the charger is damaged, please replace it in time.
- When using the robot for the first time, please charge the battery to 100%.
- If the robot is not used for a long time, please charge the robot every 2-3 months to keep the battery in good condition.
- Ensure that the power supply voltage meets the voltage marked on the charger, otherwise the charger may be damaged.
- Do not drop or collide the charger with other objects to avoid damaging the charger.
- Do not use damaged batteries. Please dispose of the battery in accordance with local regulations and do not dispose of the battery as domestic waste. Improper disposal of the battery may cause the battery to explode.

1.2 Usage instruction

- It is forbidden to clean and maintain the robot while it is turned on and running.
- It is forbidden to place flammable and explosive items on the robot.
- Do not take or place items during the movement of the robot to avoid property damage and personal injury caused by accidental collision.
- Do not push or carry the robot during the movement of the robot to avoid abnormal walking of the robot.
- Unprofessional personnel are not allowed to disassemble and repair the robot without authorization. If the robot fails, please contact the technical support engineer in time.
- When moving the robot, the maximum weight allowed by local laws or regulations for a single person cannot be exceeded. Please always keep the robot upright during the movement.
- Do not play or frolic in front of the robot while it is moving to avoid unnecessary injuries.

- The robot supports automatic obstacle avoidance. It is strictly forbidden to suddenly block the robot during high-speed driving to avoid safety accidents.
- Do not subject the robot to strong impact or vibration to avoid damage to the robot.
- Do not use strong chemicals, cleaning agents or strong detergents to clean the robot. Please use a clean, dry cloth to wipe the robot.
- If an emergency occurs during the movement of the robot, please press the emergency stop button on the top or bottom of the robot to stop the robot.
- Do not spill any liquid into the robot to avoid damage the robot.

2.Robot introduction

There are 3 kinds of robots in the X300 series:

X300B-L: a lifting robot that can perform lifting and flatbed handling, mostly used for delivering material in production line (shelf mode/material cart mode), process turnover and other processes.

X300B: a flatbed handling robot, mainly used for material transportation and process turnover of material boxes and packaging boxes.

X300T: a flatbed handling robot with multi-layer trays, mostly used for CPU, chip, wafer and other object distribution in 3C electronics and semiconductor industries.

X300B series robots are intelligently used in factories, warehouse automation fields or other handling scenarios, with autonomous navigation, automatic obstacle avoidance, autonomous handling, automatic charging and other functions. Different from the heavy system, heavy network and heavy deployment solutions of traditional industrial AMR, X300B series robots pay more attention to robot intelligence, creating light deployment and easy-to-operate robot products, which can be used without deploying servers, LAN environments and beacons, greatly saving deployment and operating costs, and helping enterprises reduce costs and increase efficiency.

2.1 Graphic description

X300B-L/300B diagram：

X300T diagram：

The X300B series robots are equipped with two diagonal Lidars to achieve 360° horizontal detection and perception capabilities, and two sets of depth cameras to detect the high, medium and low 3D range detection in front to enhance the detection capability of obstacles and help the robot perceive and avoid obstacles. At the same time, the robot is equipped with three visual sensors for VSLAM and AI recognition, which are integrated with laser SLAM to realize beacon-free deployment navigation solutions.

2.2 Parameter list

Models		**X300B-L** **With lifting**	**X300B** **Without lifting**	**X300T** **With tray**
Basic parameter	Size（L*W*H）	852*506*1270	852*506*1270	852*506*1270
	Chassis height	270mm	270mm	270mm

	Rotating diameter	950mm		
	Ground clearance of chassis	25mm		
	Weight	≈120kg	≈120kg	≈120k
	Maximum payload	300KG	240KG	
	Lifting height	55mm	\	\
	Lifting time	10s	\	\
	Maximum speed (Unload)	1.0m/s		
	Maximum speed (Full-load)	0.7m/s	0.7m/s	0.7m/s
	Minimum passage width (unload)	70cm		
	Turning width (Unload)	125cm		
	Minimum passage width with shelf	110cm	\	\
	Minimum turning width with shelf	135cm	\	\
	Max climb slope	5°		
	Maximum obstacle passability	20mm		
	Maximum gap passability	35mm		
	Robot operating temperature	0~45℃		
	Battery operating temperature	0~45℃		
	Navigation accuracy	±20mm，2°		
	End positioning accuracy	±10mm，1°		
Technical Parameter	Drive mode	Two-wheel differential drive		
	Navigation method	Laser SLAM+VSLAM+vision-assisted navigation		
	Map resolution	5cm (default); configurable 2cm resolution		
	Max mapping area per time	20000 ㎡		

	End positioning method	Ground QR code method, marker method, reflective column method		
	Laser parameters (Single)	WLR-716-Mini, range 0.1-40m (90% reflection), accuracy ±25mm		
	Depth camera parameters (single)	Upper depth camera FOV: H91°V62°± 3° Detection distance: 3m		
	RGB camera parameters	OV9782, Resolution 1920*1080 Global exposure FOV:D124 °H115°V80 °		
	Algorithm board parameters	RK3588 8 cores 8 threads max 2.1GHz		
	Interactive screen parameters	10.1-inch RK3288 Android screen Quad-core Cortex-A17 with a maximum frequency of 1.8GHz Memory/storage: 2GB/8GB Resolution: 1280*800		
	Battery type	Lithium iron phosphate battery 24V30Ah		
	Battery cycle life	Charge and discharge more than 2000 times, SOC≥ 80%		
	Charging method	Manual charging/automatic charging		
	Operating time	18h (Unload), 14.5 h (Load), 13.5h (Full load)		
	Charging time	0-100% 3.5h		
		30~80% 2h		
	Communication method	4G/Wifi/Lora(Optional)/Esp32		
	Remote control	Optional		
	Obstacle detection method	High object detection, low object detection, cliff detection		
	Obstacle detection height	Main LiDAR obstacle avoidance: 200mm Vision/low object obstacle avoidance: ≥50mm		
	Safety circuit	Not support		
	Status indicator	Top warning light, front and rear side light strips		
	Emergency stop button	3	3	3
	Anti-collision strip	Omni-directional anti-collision strip		
	Overload & unbalanced load	No		

	detection			
	Waterproof	IP21		
Button Peripheral API	Power circuit breaker button	Inside the hinge on the right rear of the robot (please refer to the manual for details)		
	Robot power button	In front of the robot (refer to the manual for details)		
	Emergency stop button	3pcs, on the top of the robot, the front and the back of the chassis (please refer to the manual for details)		
	Manual/auto button	1pc on the right side of the column, 1pc on the front side of the chassis (please refer to the manual for details)		
	SIM card slot	The rear side of the robot chassis (please refer to the manual for details)		
	Diagnostic & data transmission interface	The rear side of the robot chassis (please refer to the manual for details)		
	Barcode scanner reserved interface	Android screen rear end column (please refer to the manual for details)		
	Power supply interface	Inside the robot cover		
	Reserved expansion board communication interface	Inside the robot cover		
	Reserved mechanical fixing position	Robot bearing surface		
Shelf	Legs of shelf	Square tube ≥30mm (preferably silver, white matte material, etc.)	\	\
	Minimum shelf size	700mm (entry surface) * 400mm * 300mm (shelf bottom height)	\	\
	Maximum shelf size	900mm (entry surface) * 800mm * 300mm (shelf bottom height)	\	\
	Standard shelf size	700mm*700mm*300 mm（shelf bottom height）	\	\

	Shelf bottom height	300mm±10mm	\	\
	Max shelf height	1500mm	\	\
	Shelf center of gravity height	900mm	\	\
	Load center of gravity offset	60mm	\	\
	Distance on both sides of the shelf	20cm	\	\
	Distance behind the shelf	20cm	\	\
	Distance between two shelves	20cm	\	\
Environment	Indoor environment	Indoor. Avoid dust, water, oil smoke or other pollutants; Keep away from flammable, explosive or corrosive gases and liquids.		
	Humidity	10~95%, no condensation		
	Altitude	Below 2000 meters above sea level		
	Floor cleaning	Keep floors clean and free from dust, particles, oil or liquids.		
	Floor material	The floor should be made of static-dissipating materials.		
	Ground bearing	Hard ground, bearing capacity 1.5 tons/square meter, allowable pressure greater than 1.5MPa.		

2.3 Main function

The X300B-L robot is equipped with a mapping deployment tool. Through the mapping software, the map, the points and task routes can be quickly created before performing the tasks. The robot can navigate autonomously, effectively detect obstacles and avoid them in real time. It supports cruise tasks, point-to-point tasks, and multi-point tasks. The point actions and duration can be set. Human-robot interaction can be performed through the touch screen and the quick release button. Its main functions are as follows.

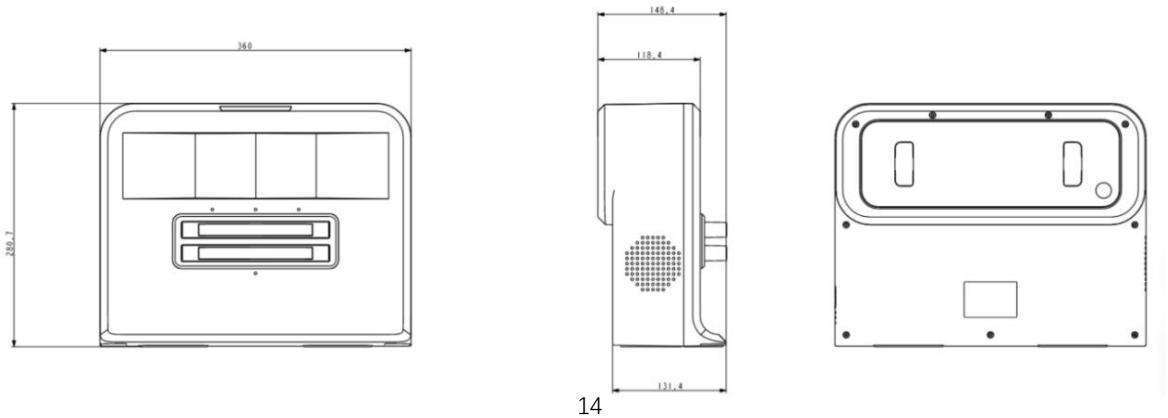
No.	Function	Function name	Function Introduction
1	Autonomous navigation	Laser SLAM & vision navigation	The robot is equipped with 2 main lidars, 2 depth cameras, 2~3 RGB visual cameras, IMU, encoder, and realizes the laser SLAM and visual SLAM fusion solution through multi-sensor fusion technology. The full-scene

			perception can adapt to the scene, and effectively navigate and avoid obstacles.
2	shelf/cart docking capability	shelf/cart docking	The X300B-L (lifting type) robot can use laser docking to detect shelves and dock shelves. Even when the shelves are offset (±30cm, ±25°), it can still effectively detect and dynamically plan the path to dock shelves. The powerful sensing and detection capabilities and path planning capabilities ensure the handling of shelves/carts.
3	Lifting delivery	Lifting delivery	***Only for model X300B-L (Lifting version)** The robot can lift the shelf/cart through the lifting mechanism to delivery. Lifting stroke is 60mm.
4	Delivery form	Flatbed delivery	Without using a carrier, directly put the materials on the robot's panel for delivery.
		Shelf/cart delivery	With shelves/carts, the robot docks, lifts shelves, and performs handling. Shelf size: Meet design requirements, any shelf within the specified range is acceptable.
5	Delivery task type	Single point delivery	The robot can perform flatbed handling/lifting tasks from point A to point B. Note: If the lifting model is used as a flatbed model, please select "flatbed handling" in the APP. For example, the raw materials are delivered to the thread head and the production process flow.
		Multi-point delivery	The robot can perform flatbed handling/lifting tasks for multiple work points at a time. Note: If the lifting model is used as a flatbed model, please select "flatbed handling" in the APP. For example, the raw materials are delivered to multiple points on the production line.
		Multi pick-up points	The robot can pick up the goods at multiple points/stations and finally return to the receiving point.
		Cruise task	Set a circular route, the robot delivers in sequence, and can set the duration of stay or stay forever (the staff taps

			the "one-click release" button or clicks "next" on the screen to continue the next task).
		Quick task	Quickly set a temporary task to execute, this task will not be saved to the task list.
		Custom task	Freely configure the flatbed/lifting handling task, this task will be saved in the "task list" after setting successful, , and the staff can select the task to execute in the task list.
6	Other functions	No-load and load switch model	The robot adapts to different traffic parameters when no-load or load to ensure safe passage.
		shelf offset detection	Even if the shelf is offset to a certain extent (horizontally ±30cm, forward 10cm, rear 30cm, ±25°), the robot can still detect the shelves and perform docking, lifting and transportation.
		Active safety	The robot has powerful perception and detection capabilities and can effectively detect, avoid obstacles. When the robot's road is blocked, it will re-plan the path.
		Automatic charging	When the robot is low power, it can return to the charging station to recharge. The threshold of low power can be set.

2.4 Charging pile

The robot supports manual charging and automatic charging. The dimensions and parameters of the charging pile are as follows:



Item	Parameter
Size（L*W*H）	360mm*148.4mm*280.7mm
Weight	≈2.8kg
Operating temperature	0~45℃
Operating humidity	30%~70%
Input power	100~240VAC 50/60Hz
Output power	29.4VDC 10A MAX
Current protection	Built-in 20A filter protection fuse
Short circuit protection	Withstands continuous short circuit and automatic recovery capabilities.

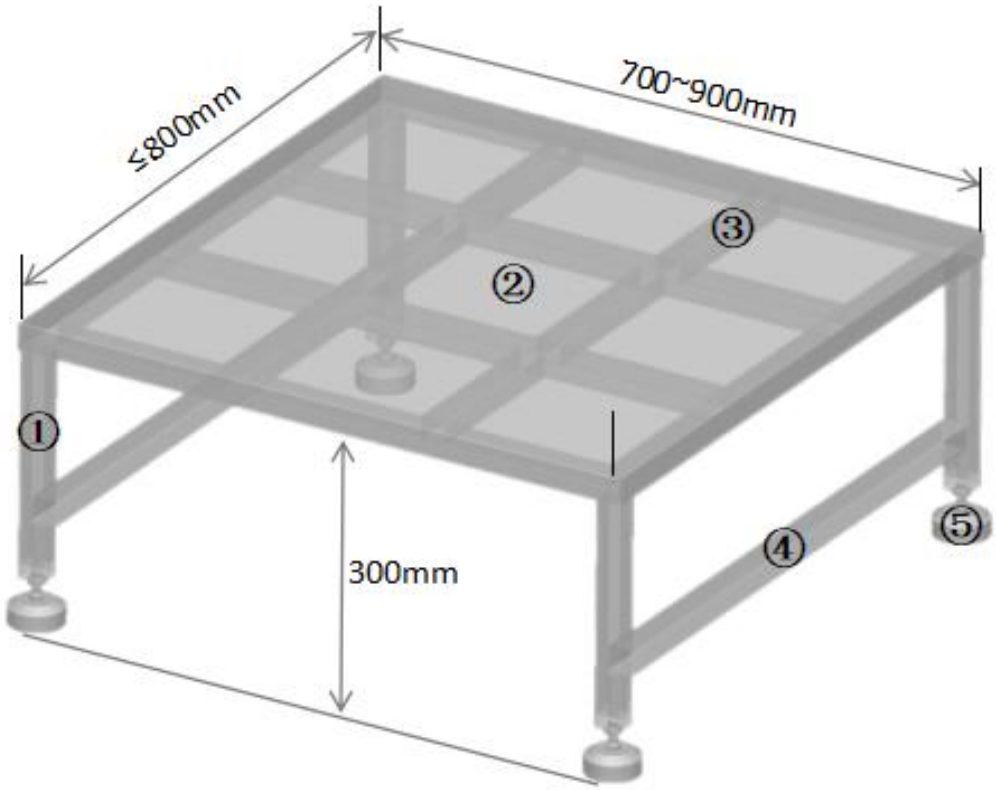
The indicator light description of charging pile

Process	Condition	Indicator status	Description
Power on	Charging pile power on	Red light is on	The charging pile is powered on normally.
Power off	Charging pile power off	Off	Charging pile is powered off.
Fast charging	Normal charging	Green light slowly flash (breathing flash)	Indicates that the robot is charging.
	Low current charging	Green light is on	Indicates that the robot is trickle charging.
Abnormal status	Charging pile works abnormally	Red light flash	The charging pad is abnormal

2.5 Shelf/cart

i300B-L can be used with shelf/cart:

- Current standard shelf size: 700mm*700mm*300mm (height of shelf bottom);
- Within the shelf size range, customers can customize the shelf size (material, bracket leg diameter, shelf bottom height must comply with iBEN ROBOT X300B-L shelf design requirements, shelf size diagram needs to be confirmed by engineer);
- If the shelf size exceeds the product definition, you need to contact engineer for confirmation;
- Shelves/carts can be designed based on the specifications and loading requirements of the customer's goods. In principle, the size of the top loading does not exceed the size of the shelf legs. If there are special-shaped shelves/carts (the size of the top loading of the shelf car is larger than the size of the shelf legs), you need to contact engineer for separate configuration.



Shelf/cart diagram		
No.	**Name**	**Description**
1	Shelf columns	The column is square, and the width is ≥30mm. The column surface is preferably white or silver matte.

2	Shelf load-bearing surface	Shelf load-bearing panels, customers can choose bracket type or whole panel; Recommended height of the bottom of the load-bearing surface from the ground: 300mm. Height range of the load-bearing surface from the ground: 290mm~310mm.
3	Shelf load-bearing beam	The crossbeam on the load-bearing surface is a "well" structure, ensuring that the front, middle and rear of the shelf crossbeam are in contact with the robot's load-bearing surface (to avoid skew caused by the crossbeam being unloaded).
4	Auxiliary support beam	The height of the beam between the shelf legs from the ground needs to be ≤150mm, and the width of the beam ≥30mm.
5	Stand foot plate (foot pad)	The foot plate of the bracket supports versions with or without adjustable foot pads. The picture shows a schematic diagram with adjustable foot pads. Note: If it is a material cart, it has universal wheels.

Shelf specifications			
Item	**shelf size**	**Cart size**	**Explanation**
The width of shelf leg	30mm(default value)		If there is a special size, it needs to be confirmed. The surface of the column is preferably white or silver matte.
Minimum size	700mm*400mm	800mm*400mm	700mm for robot to enter
Maximum size	900mm*800mm		900mm for robot to enter. If it exceeds the scope, R&D evaluation is required
Standard size	700mm*700mm	800mm*700mm	800mm for robot to enter
Assembly height of wheel	/	≤120mm	

Bottom height	300mm	Robot chassis is 270mm. Lifting stroke is 60mm. Recommended shelf bottom dimensions: 300mm Range: 290mm~310mm
Maximum height	1500mm	
Placement requirements of shelf/Cart	Reserved 150mm on both sides Reserved 200mm on rear side Reserved 1000mm on front side	
Space between multi shelf/cart	150mm	

2.6 Industrial application hardware support

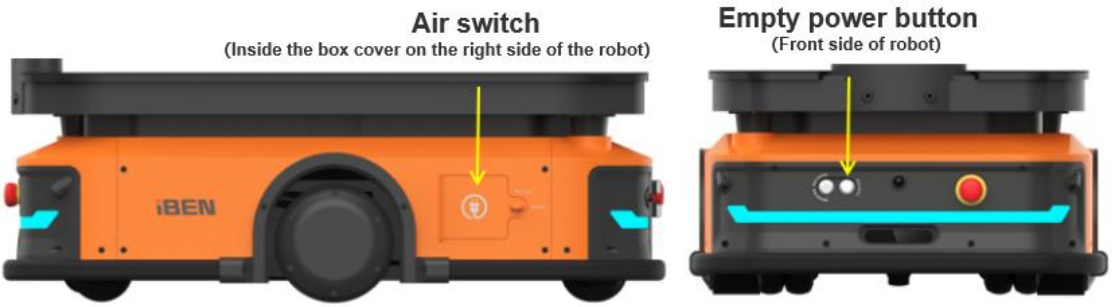
iBEN ROBOT has simultaneously adapted industrial IoT devices to help customers realize fully automatic and semi-automatic handling solutions. X300 supports the following solutions:

- Pager (WIFI version): It can be used after connecting to the factory network (Internet), supporting point-to-point calling, task group calling, and multi-pager task scheduling.
- WIFI router: Used with the pager to establish a local area network that can be connected to the Internet, and supports connecting multiple pagers within a range;
- 4G router: If the factory has no WIFI network that can be connected to the Internet, a 4G router can be used to connect the pager.
- Elevator control module: A module used for elevator control so that the robot and the elevator can be interconnected.
- Automatic door module: An automatic door module so that the robot can pass through the automatic door.

3. Robot installation and usage-APP operation

3.1 Power on and off

The robot can be turned on and off by operating the power button, which is located on the chassis of the robot.

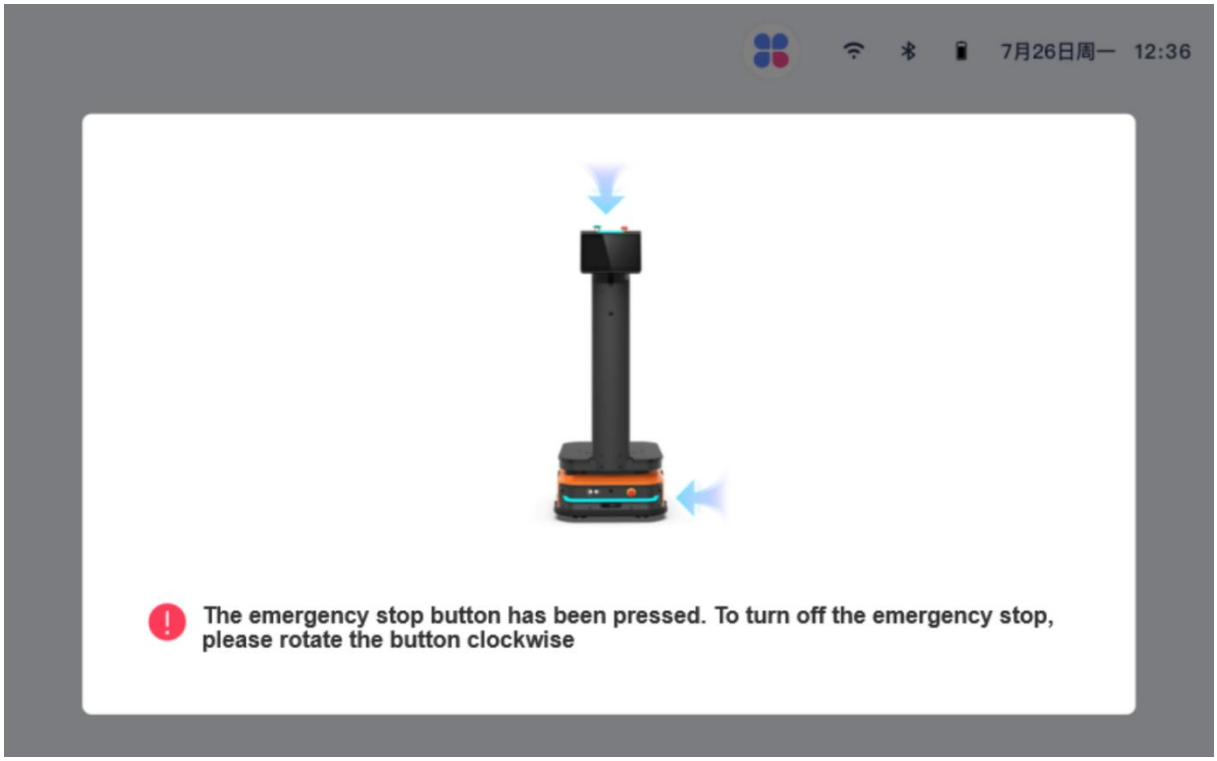


Power on: first turn on the Air switch, press the power button once, the indicator light turns blue, indicating that power on successfully.

Power off: long press the power button, the indicator light turns off, wait for the robot to shut down and then turn off the circuit breaker.

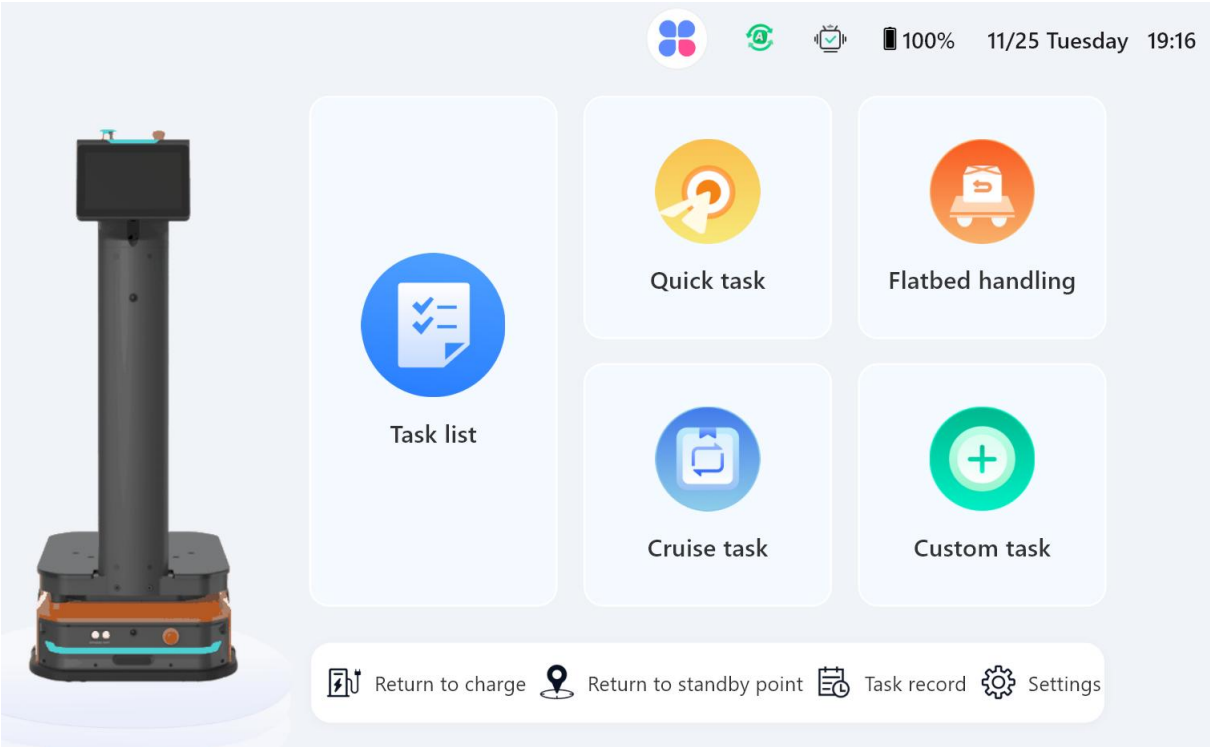
3.2 Emergency stop button

There are emergency stop buttons on the top of the robot and on the left and right sides of the chassis. In case of emergency during movement, press the emergency stop button to stop the robot. Turn the emergency stop button clockwise and follow the prompts on the interface to resume the robot's operation.



3.3 Task configuration

In order to adapt to different industrial scenarios, the X300 robots provide users with a variety of task modes. After the robot is turned on, users can choose task modes according to different needs on the home page.

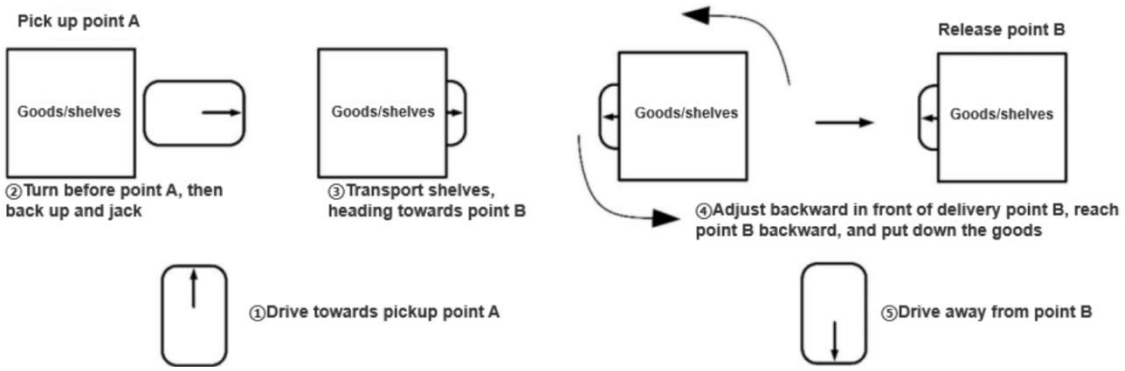


Task Modes	Description
Quick task	Users can set jack-up and jack-down points. The robot will automatically move to the jack-up point, identify the shelf, lift the goods, transport them to the destination, and automatically unload the shelf.
Cruise task	Users can set a route, and the robot will run along the route for a set number of times. It can stop at designated points on the route for a specified duration, facilitating the pickup and delivery of goods during patrol. After task setup, it will be automatically saved in the task list, making it easier to execute the task quickly.
Flatbed handling	Enables convenient configuration of single-point-to-multi-point flatbed handling tasks. This task mode does not support lifting.
Custom task	Users can set a route, and the robot will run along the route for the set number of times. It can jack-up and jack-down shelves at designated points on the route, stop at points for a specified duration. After task setup, it will be automatically saved in the task list, making it easier to execute the task

	quickly.
Task list	Allows viewing, editing, starting, and deleting the cruise tasks and custom tasks.

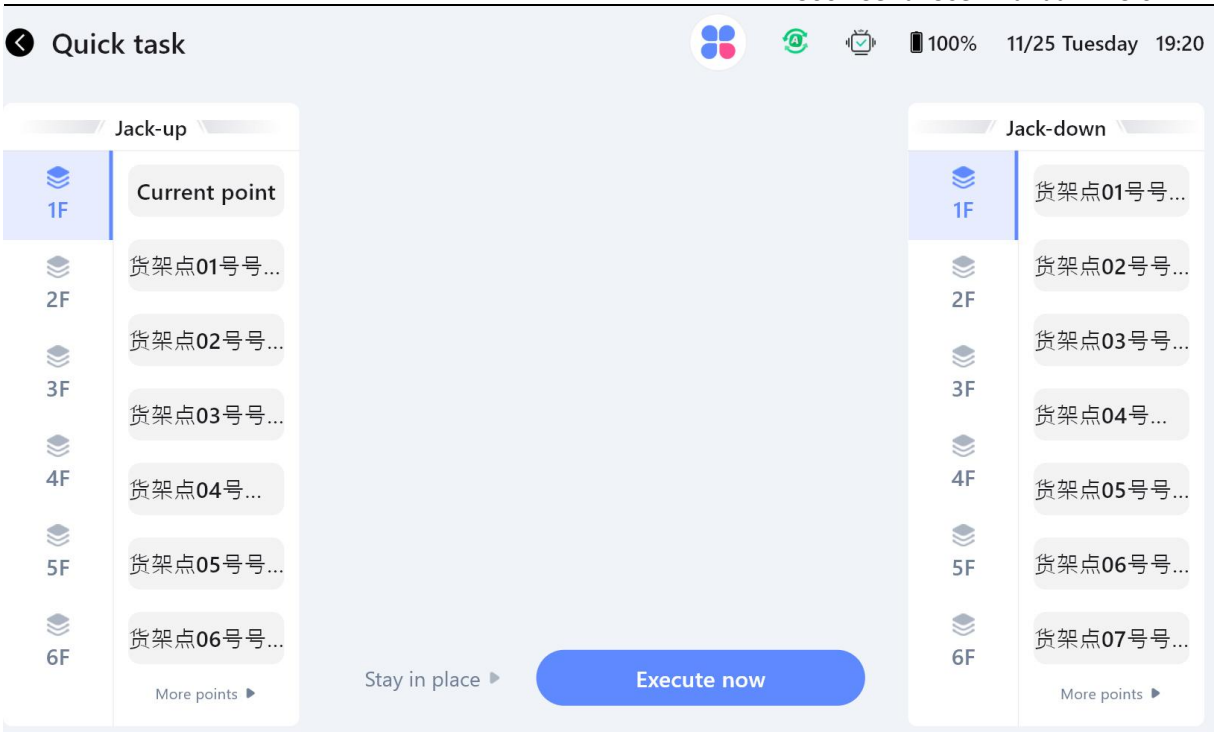
3.3.1 Quick task

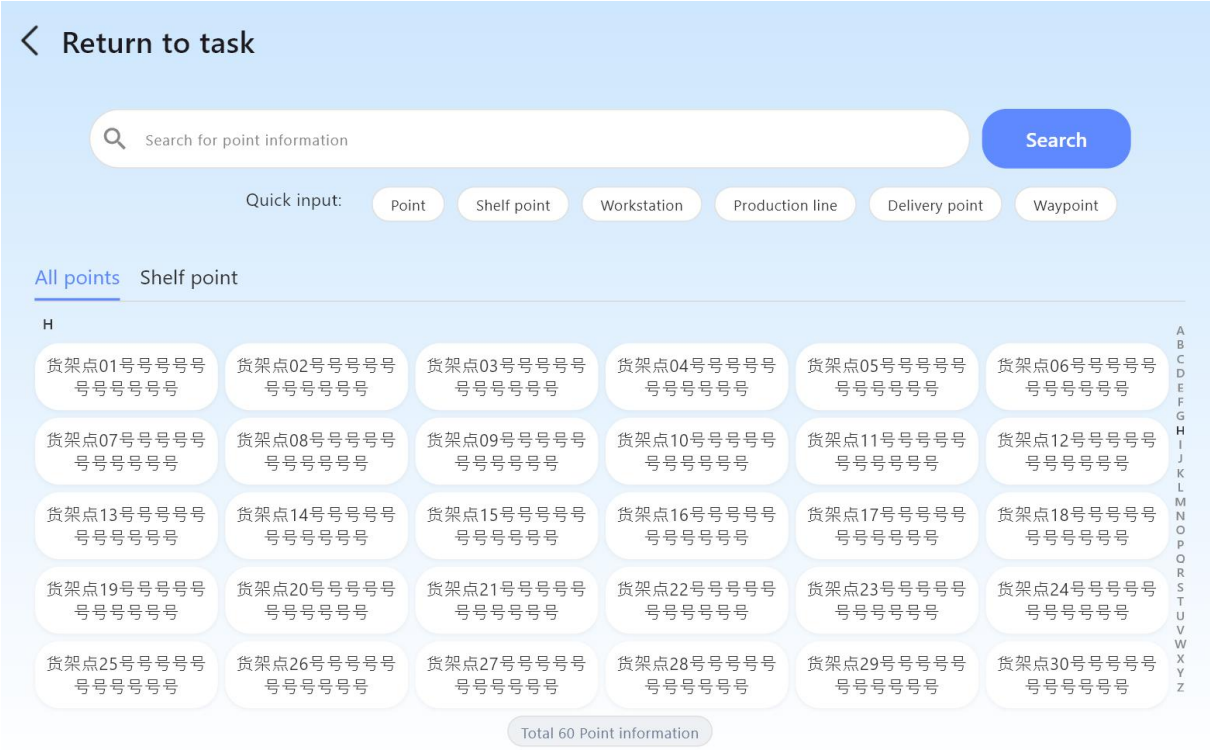
Users can set the pick-up point and the delivery point. The robot can automatically go to the pick-up point to identify the shelf and lift the goods. After transporting the goods to the destination, it will automatically jack-down the shelf.



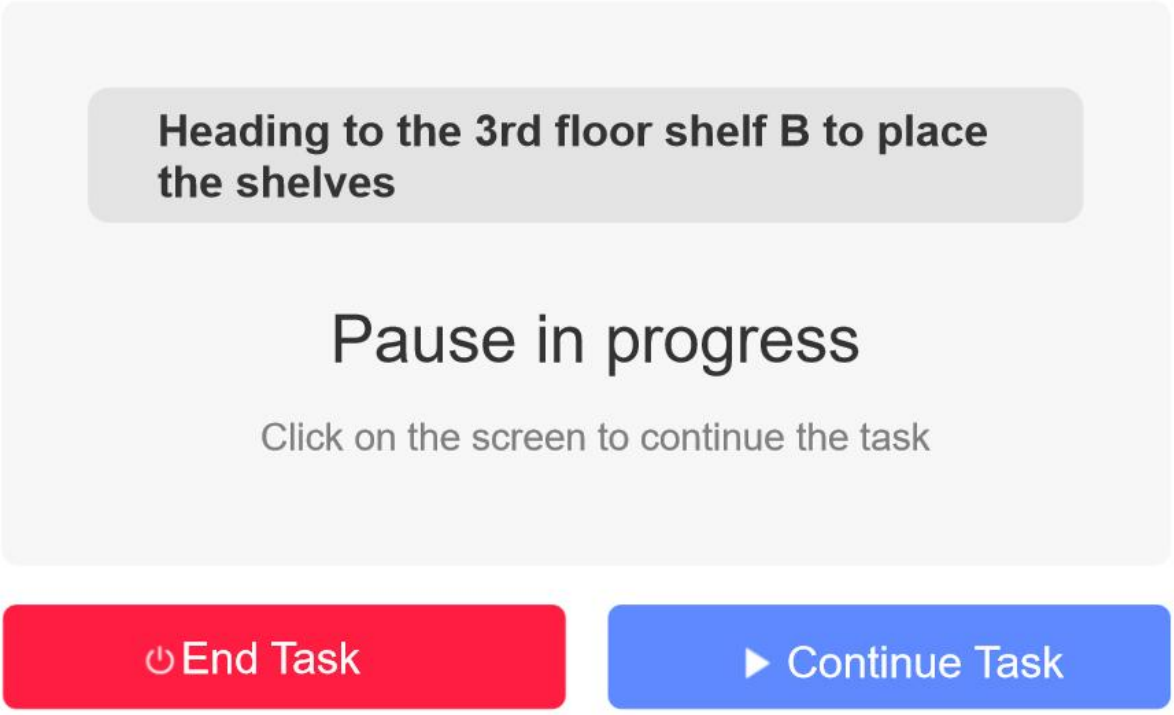
Quick task configuration process

1. Select [Quick task] on the homepage to enter the quick task interface shown in the figure below.
2. Select the point where you want to transport the shelf in the "Jack-up point" list on the left side of the interface, and select the point where you want to place the shelf in the "Jack-down point" list on the right side.
3. After selecting the transport route, you can choose the robot action after the transport is completed: "Return to the charging pile, return to the standby point, stay in place".
4. Click [Execute Now] and the robot will perform the task after the broadcast prompt.
5. Click on [More Points] to search for locations quickly.
6. The completed quick tasks can be viewed in the task record.





When the robot is in the process of lifting tasks, the user can click the screen to pause the robot task. At this time, the robot is in the paused task state. Click the screen or the [Continue task] button to perform the task continuously. Click [End task] to cancel the task.



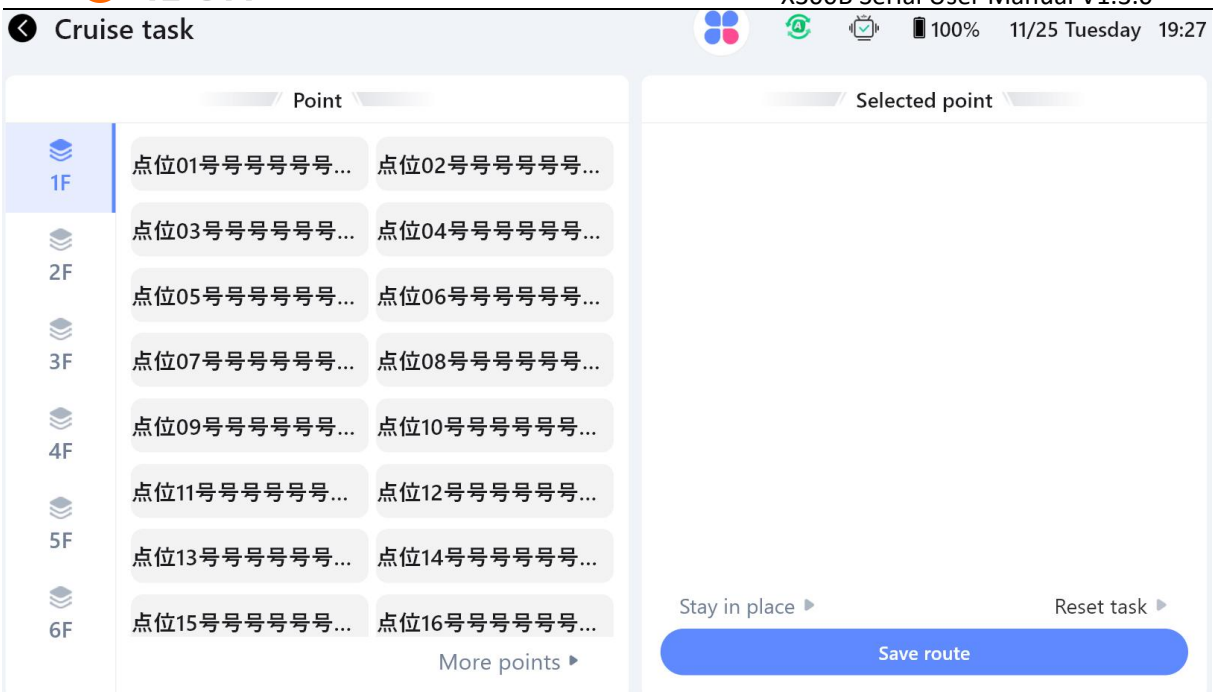
3.3.2 Cruise task

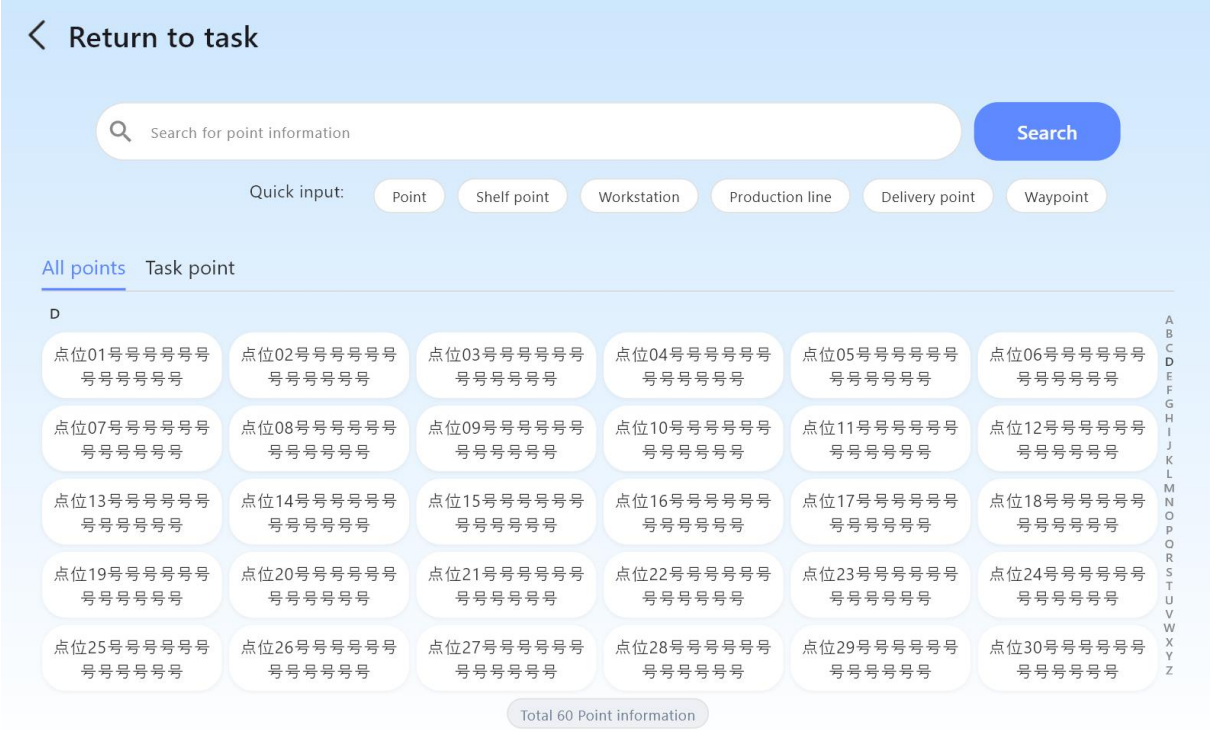
Deliver the goods to the designated point. After finishing all tasks, the robot returns to the designated location and waits for orders.

After completing the task, it will be automatically saved in the task list, making it more convenient to execute the task later. The completed cruise task can be viewed in the task record.

Patrol task configuration process

1. Select [Cruise task] on the homepage to enter the cruise task interface shown in the figure below.
2. Select the delivery point in the "Point" list on the left side of the interface.
3. The delivery route is displayed on the right, and set the duration for each point.
4. After selecting the delivery route, you can choose the robot action after the task is completed: "Return to the charging pile, return to the standby point, stay in place".
5. Click [Save Route], then enter the task name, number of cycles, scheduled task time, and whether to repeat. Cruise tasks can be viewed and executed in the task list.
6. Click [More points] to quickly search for waypoints.
7. Completed cruise tasks can be viewed in the task records.


Cruise task
100%
11/25 Tuesday 19:27
Point
Selected point
1F
2F
3F
4F
5F
6F
点位01号号号号号...
点位02号号号号号...
点位03号号号号号...
点位04号号号号号...
点位05号号号号号...
点位06号号号号号...
点位07号号号号号...
点位08号号号号号...
点位09号号号号号...
点位10号号号号号...
点位11号号号号号...
点位12号号号号号...
点位13号号号号号...
点位14号号号号号...
点位15号号号号号...
点位16号号号号号...
More points
Stay in place
Reset task
Save route


Return to task
Search for point information
Search
Quick input:
Point
Shelf point
Workstation
Production line
Delivery point
Waypoint
All points
Task point
D
点位01号号号号号号号号号号号
点位02号号号号号号号号号号号
点位03号号号号号号号号号号号
点位04号号号号号号号号号号号
点位05号号号号号号号号号号号
点位06号号号号号号号号号号号
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点位08号号号号号号号号号号号
点位09号号号号号号号号号号号
点位10号号号号号号号号号号号
点位11号号号号号号号号号号号
点位12号号号号号号号号号号号
点位13号号号号号号号号号号号
点位14号号号号号号号号号号号
点位15号号号号号号号号号号号
点位16号号号号号号号号号号号
点位17号号号号号号号号号号号
点位18号号号号号号号号号号号
点位19号号号号号号号号号号号
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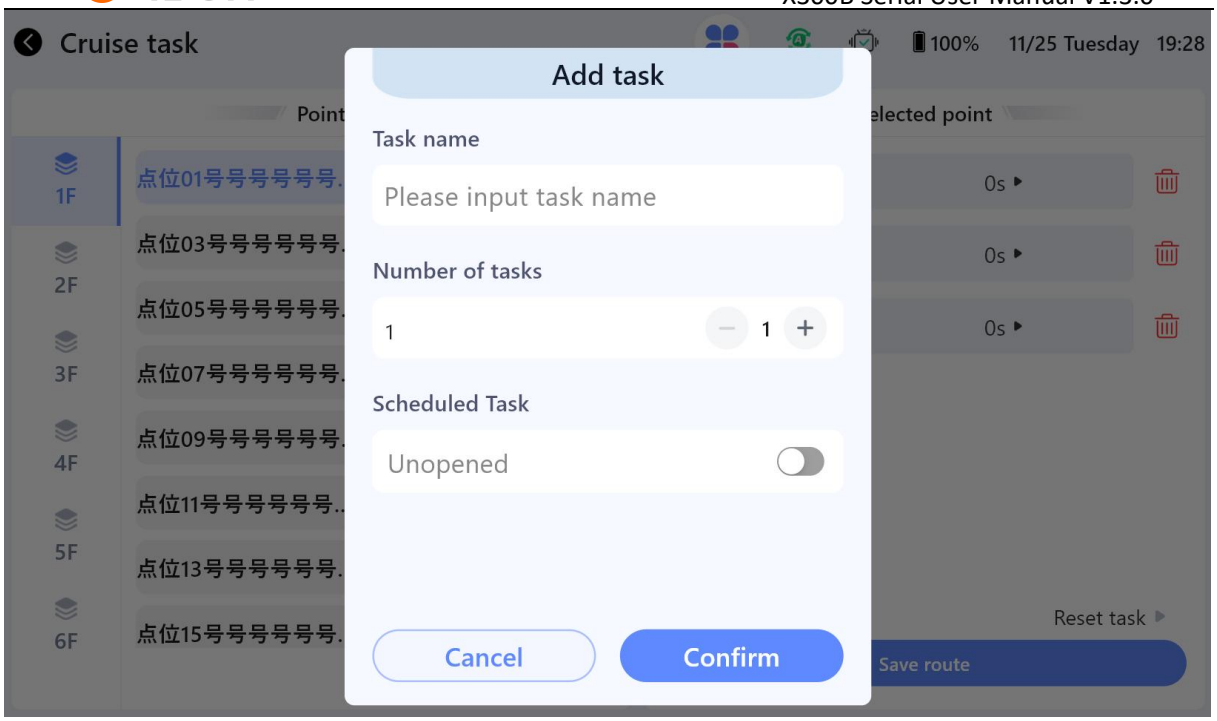
点位24号号号号号号号号号号号
点位25号号号号号号号号号号号
点位26号号号号号号号号号号号
点位27号号号号号号号号号号号
点位28号号号号号号号号号号号
点位29号号号号号号号号号号号
点位30号号号号号号号号号号号
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Total 60 Point information

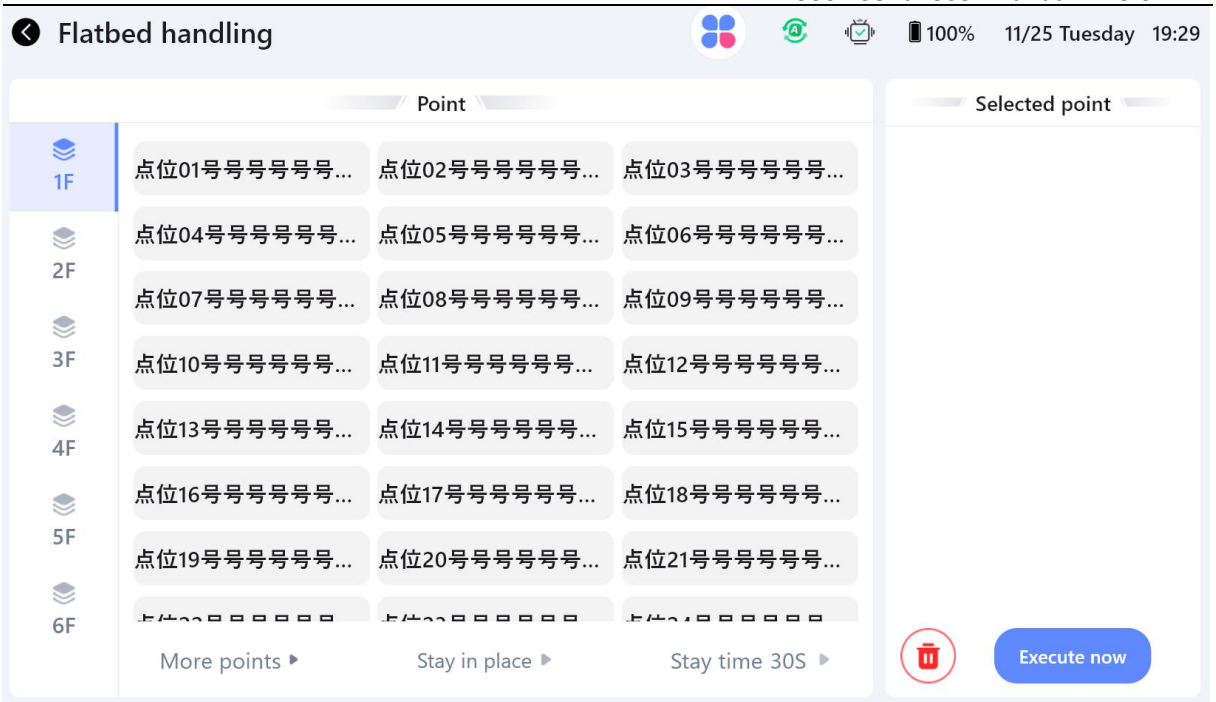


3.3.3 Flatbed handling

The robot can perform tasks at a single point or multiple points.

Task configuration process

1. Select [Quick task] on the homepage to enter the quick task interface.
2. Select points on the interface to form a route.
3. After selecting the route, you can choose the robot action after task is completed: "Return to the charging pile, return to the standby point, stay in place"
4. You can choose the speed of transportation, low-speed, medium-speed, and high-speed.
5. Click [Execute now] and the robot will perform the task.

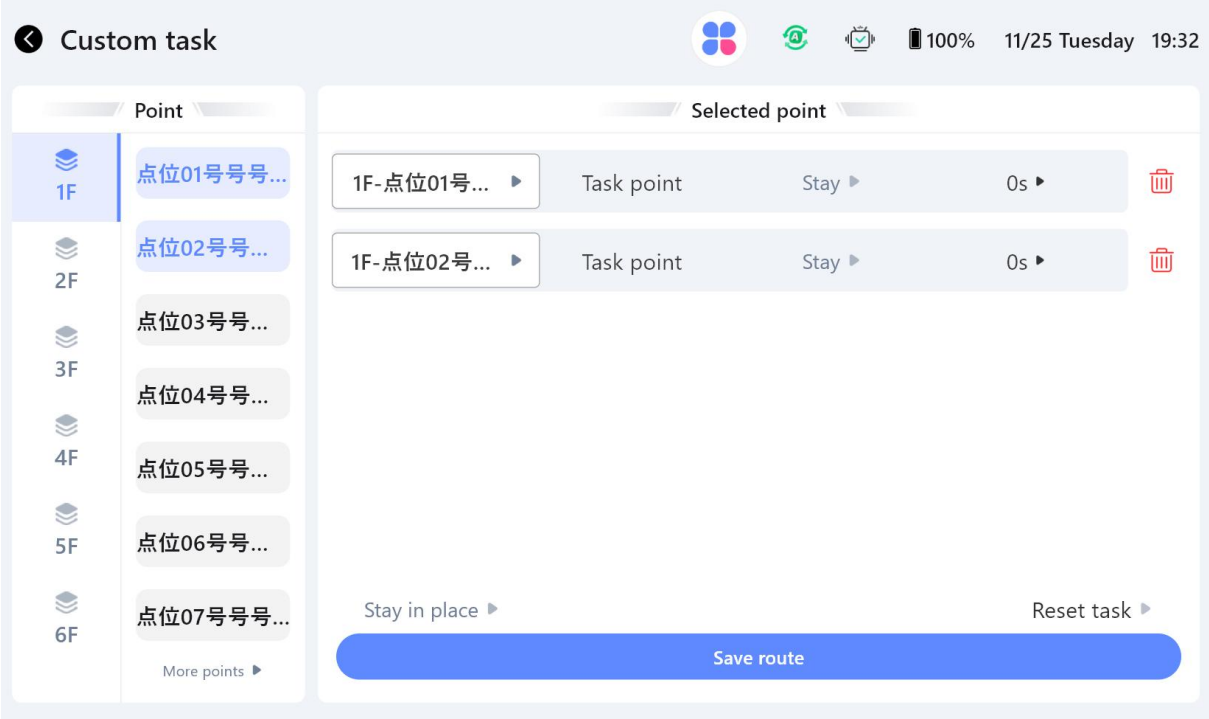


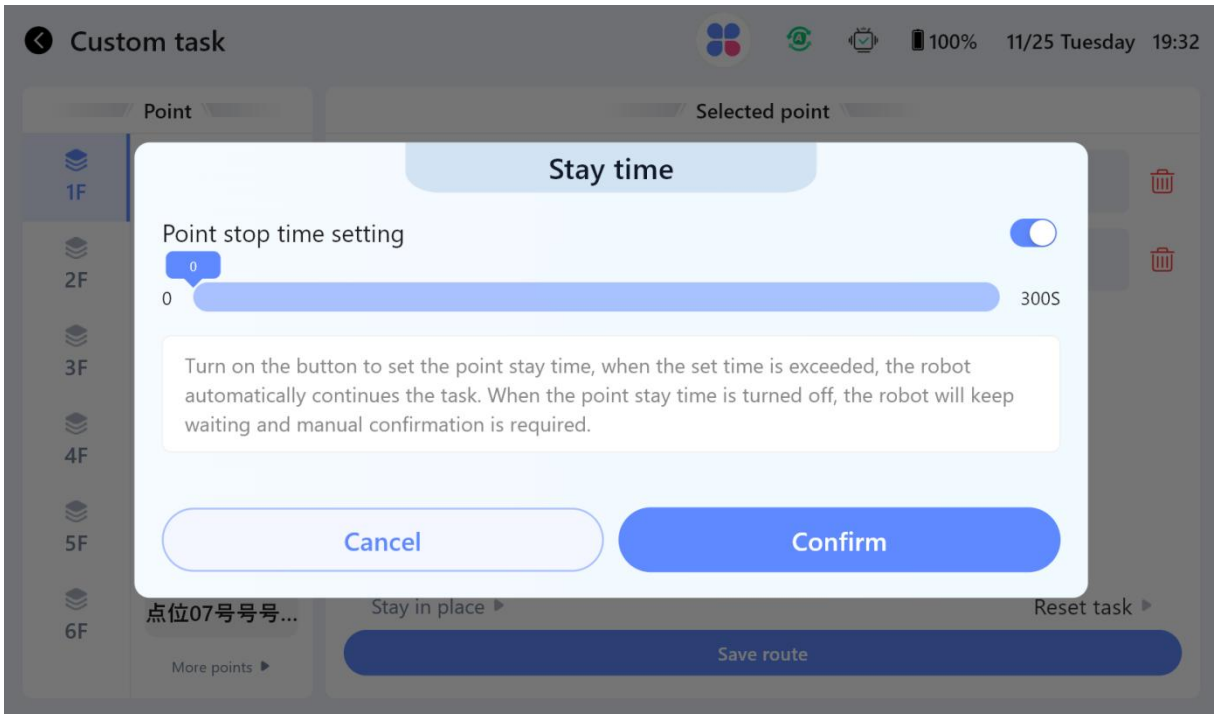
3.3.4 Custom task

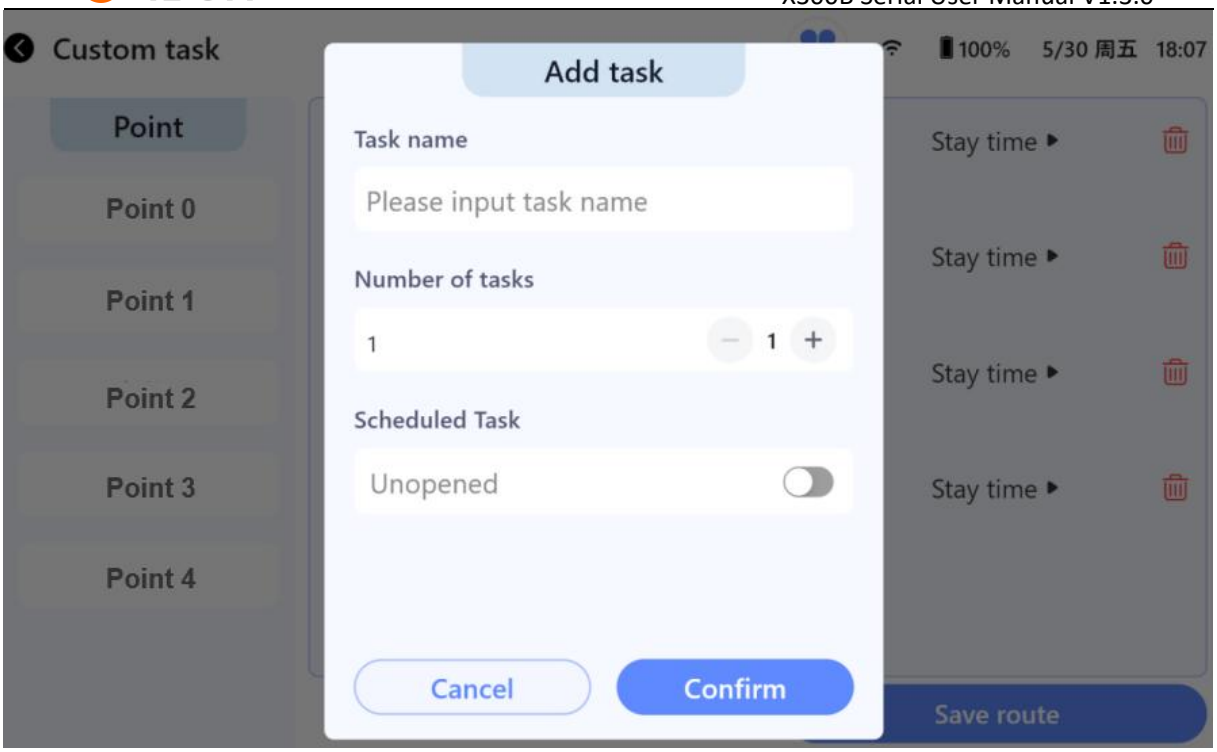
Deliver the goods to the designated point. After finishing all tasks, the robot returns to the designated location and waits for orders.

Task configuration process

1. On the homepage, select [Custom Task] to enter the custom task interface shown in the figure below.
2. In the "Points" list on the left side of the interface, select the delivery waypoints.
3. The delivery route will be displayed on the right side, and you can configure the point actions (lifting, placing, staying) and the dwell time at each waypoint.
4. After selecting the delivery route, you can choose the robot's action after completing the handling: "Return to Charging Pile, Return to Smart Charging Pile, Return to Standby Point, Stay in Place".
5. Click [Save Route], then enter the task name, number of cycles, scheduled task time, and whether to repeat. Configured tasks can be viewed and executed in the task list.
6. Click [More points] to quickly search for waypoints.
7. Completed custom tasks can be viewed in the task records.

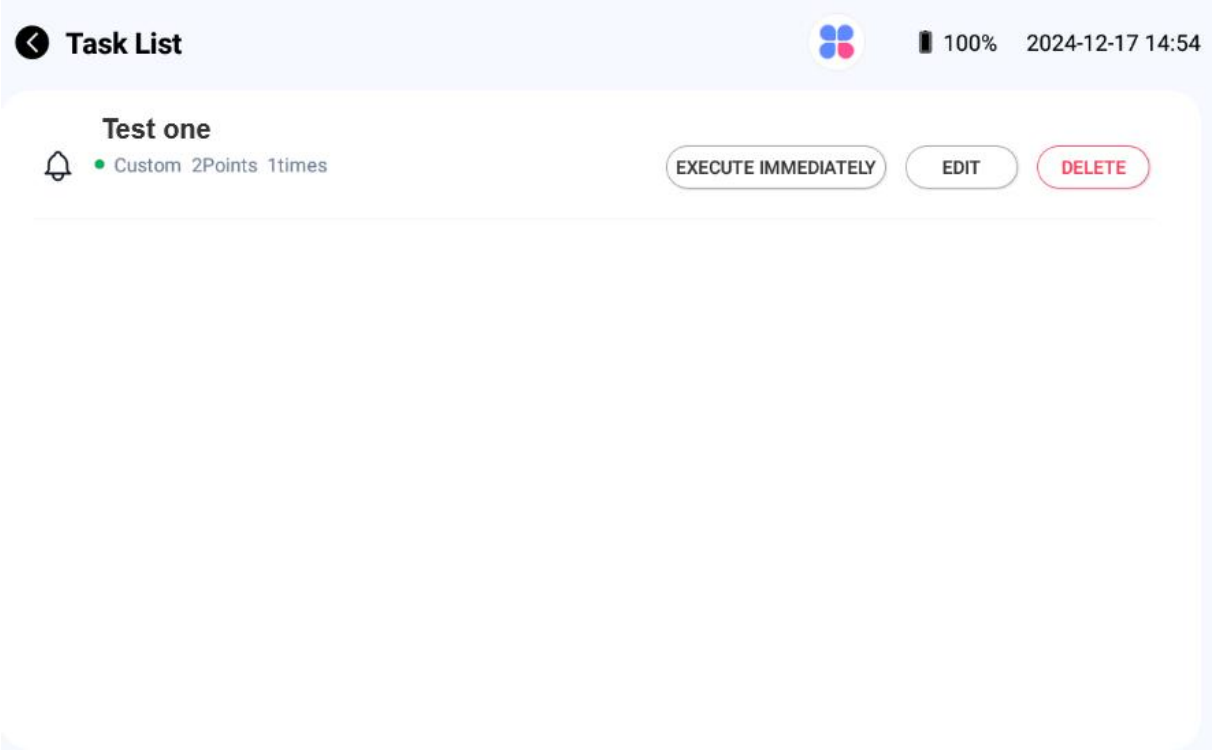

Custom task
100%
11/25 Tuesday 19:32
Point
Selected point
1F
2F
3F
4F
5F
6F
点位01号号号...
点位02号号...
点位03号号...
点位04号号...
点位05号号...
点位06号号...
点位07号号号...
More points
1F-点位01号...
Task point
Stay
0s
1F-点位02号...
Task point
Stay
0s
Stay in place
Reset task
Save route


Custom task
100%
11/25 Tuesday 19:32
Point
Selected point
Stay time
Point stop time setting
0
0
300S
Turn on the button to set the point stay time, when the set time is exceeded, the robot automatically continues the task. When the point stay time is turned off, the robot will keep waiting and manual confirmation is required.
Cancel
Confirm
点位07号号号...
More points
Stay in place
Reset task
Save route



3.4 Task list

The setup cruised tasks and custom tasks will be displayed in the task list. This interface can be used to manage the task list.



Function description of task list buttons

Button	Function description
Execute immediately	Click to make the robot execute the task immediately.
Edit	Click to edit task configurations.
Delete	Delete the task.

3.5 Return to charging pile

Click [Return to charging pile], and the robot will immediately return to the charging pile.

3.6 Return to standby point

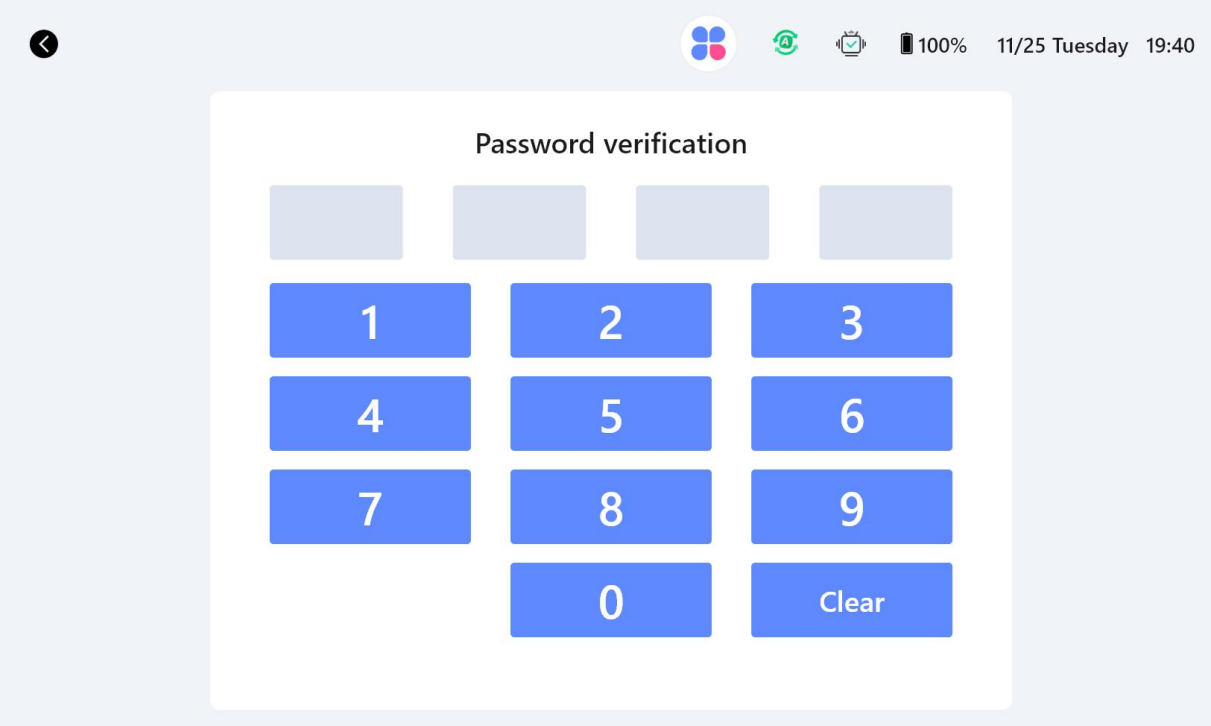
Click [Return to standby point], and the robot will immediately return to the standby point.

3.7 Task Records

It counts the total number of tasks and the number of tasks for the current day, supporting task filtering by status, type, source, and date range. It can display detailed task information, including task type, number of points, start and end time, task status (e.g., Completed, Cancelled, Failed), and task duration, facilitating the review and management of historical tasks.

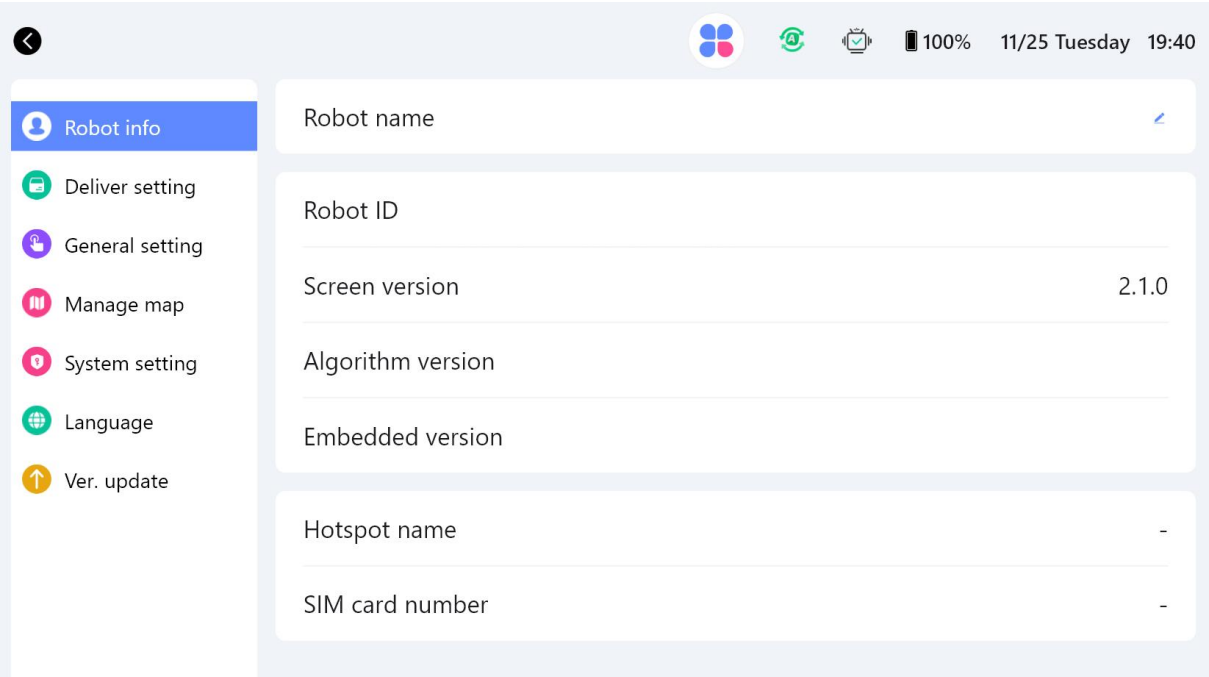
3.8 Settings

There is a [Settings] button on the homepage. Click it to enter the robot's settings interface. Please enter the default password of **6666**.



3.8.1 Robot information

You can view the basic information of the robot, such as the robot name, number, interactive screen version, algorithm version, embedded version, hotspot name, SIM card number, etc.



3.8.2 Deliver setting

Self Selection: You can set standard container dimensions (700mm × 700mm) or customize the container size.

Broadcast Settings: You can configure the specific content of task broadcast messages and low-battery reminders. Additionally, you can turn background music on or off via a toggle switch.

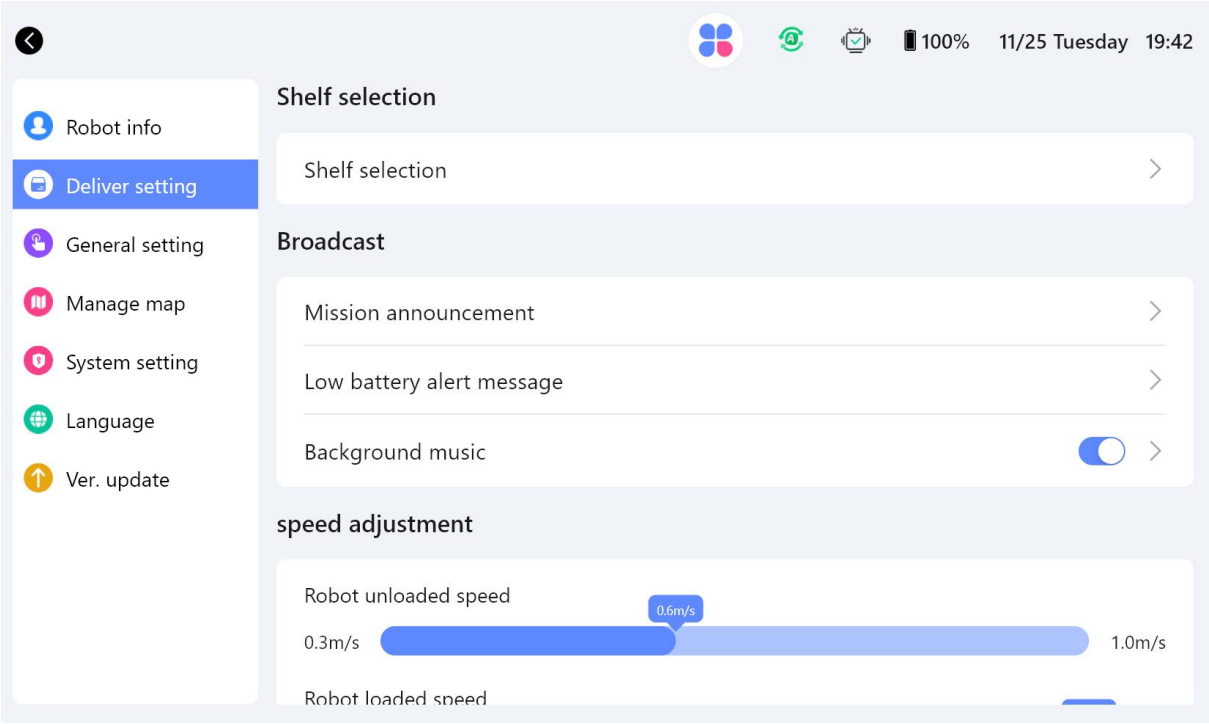
Speed Adjustment: Adjust the robot's speed when unloaded and loaded separately using sliders to meet the speed requirements of different operating scenarios.

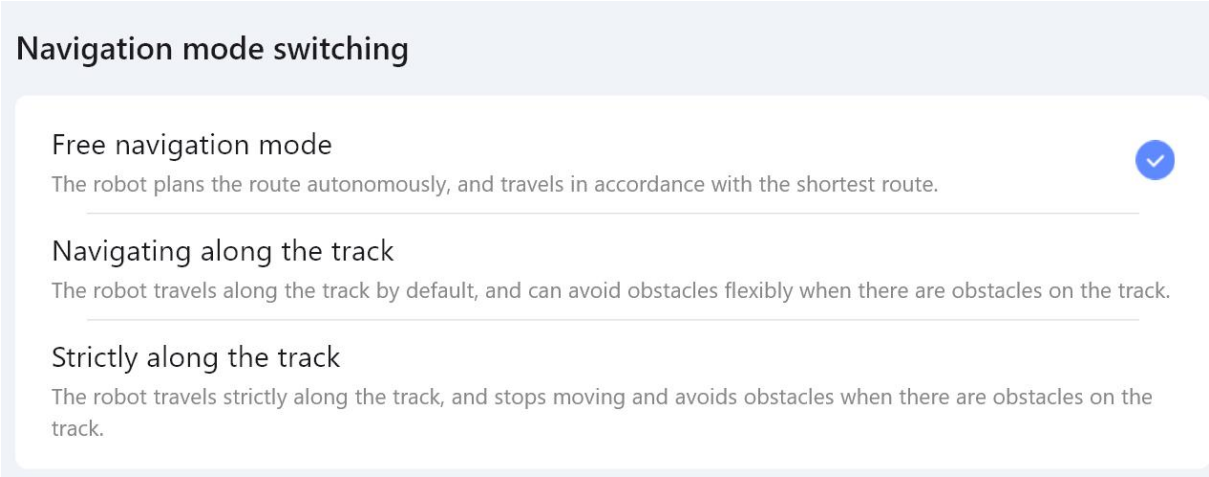
Navigation Mode Switching:

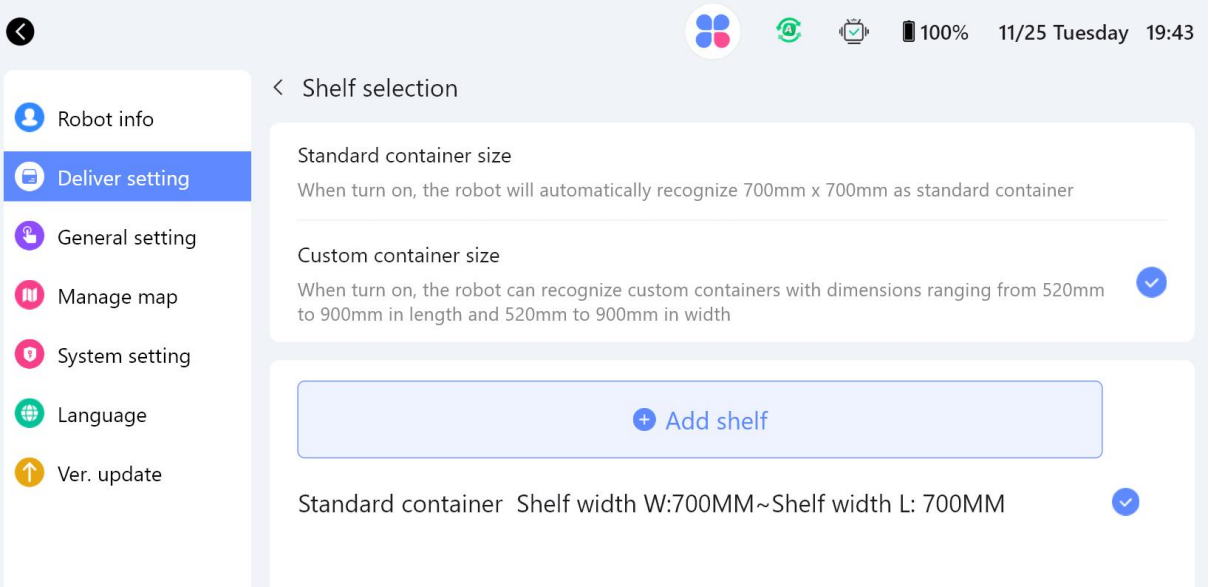
Free Navigation Mode: The robot autonomously plans its route and travels via the shortest path.

Navigating along the track: The robot travels along default tracks and flexibly avoids obstacles on the track.

Strictly along the track: The robot travels strictly along the track and stops moving if there are obstacles on the track.







3.8.3 General Settings

Positioning settings

It provides the "Pile Relocation" function for configuring the location of charging piles; it also allows configuring "Initial point positioning" for positioning the initial point in scenarios where there is no charging pile on the floor.

Charging pile settings

You can select a charging pile (ordinary charging pile) or a smart charging pile (smart scheduling is enabled by default once selected).

Elevator setup

You can set elevators to be enabled/disabled.

Collision sensor

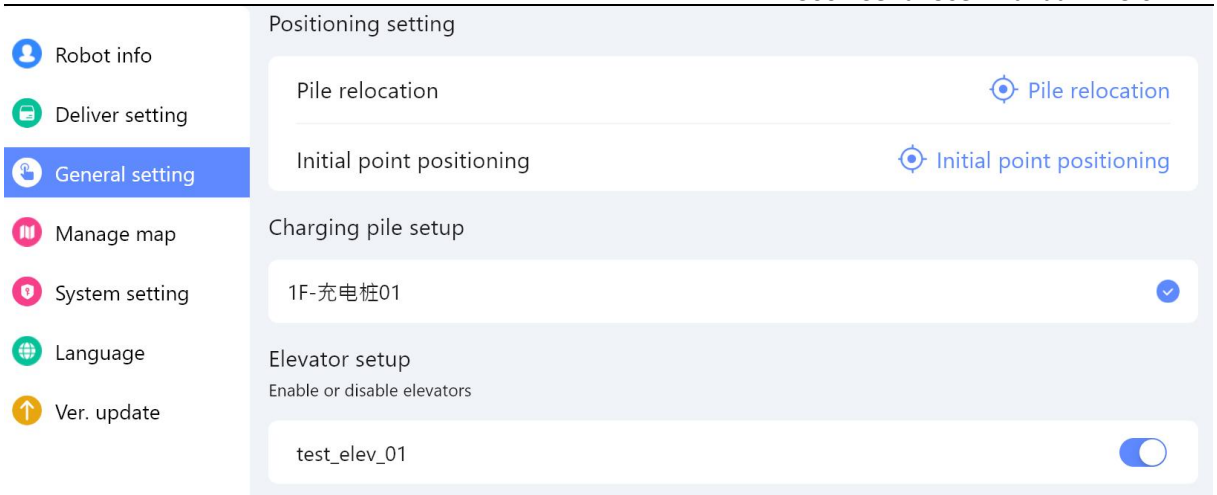
When enabled, if the robot triggers a collision, it will stop moving immediately. The collision status must be manually cleared before the robot can resume task execution.

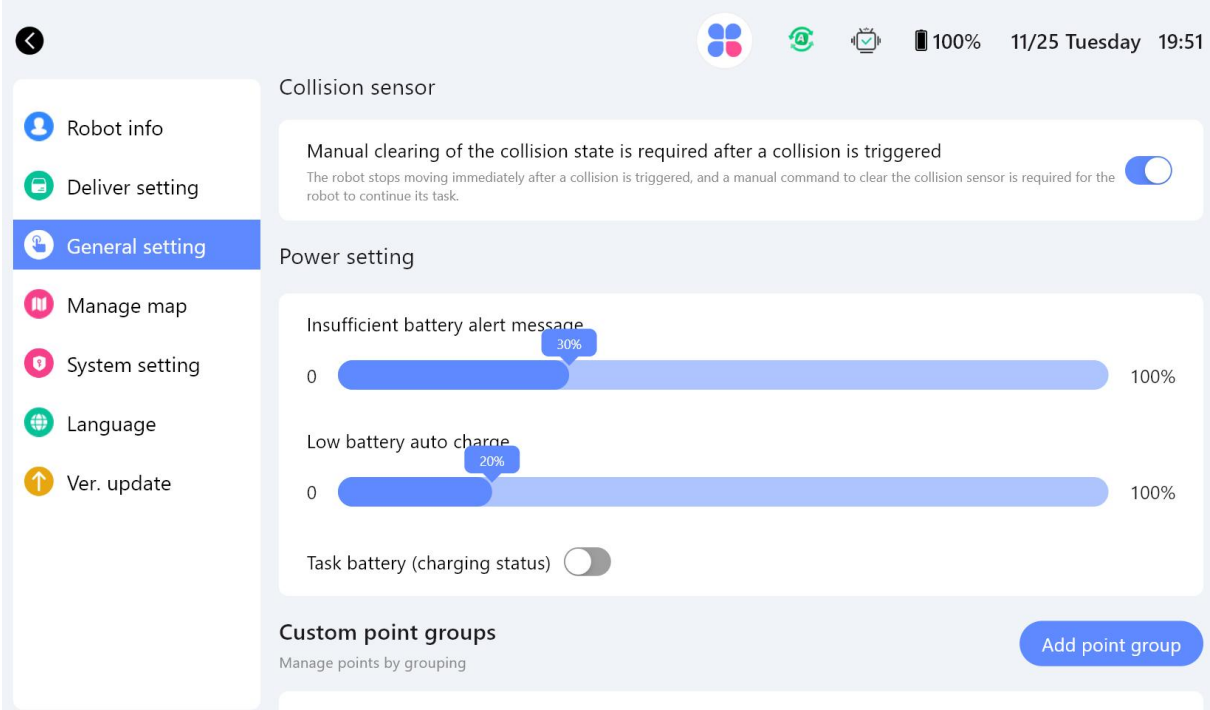
Power settings

Adjust the low-power reminder and automatic low-power charging of the robot separately using sliders to meet the power requirements of different operating scenarios; you can configure the switch for task-execution power. When enabled, the power for task execution must be higher than the threshold for automatic low-power charging.

Custom point groups

Click [Add point Group], enter the name of the point group and select the points under this group. After successful saving, a piece of group information will be displayed, and you can perform [Edit] and [Delete] operations on this group information.





3.8.4 Map Management

The Map Management is used for centralized management of maps, supporting operations such as adding new maps, switching maps, editing maps, deleting maps, uploading to the cloud, and floor management. It helps robots operate efficiently in different work areas and on different floors.

Add new map

Click the [Add New Map] button to create a new map, which is used for robots to perform delivery tasks in new work areas.

Map thumbnail

Intuitively displays the outline of the map for quick identification.

Map name

Naming of maps for different work areas.

Modification time

Displays the time when the map was last modified (e.g., "2025-10-31 10:19:32"), facilitating traceability of map update records.

Floor information

Labels the number of floors included in the map and the specific floors (e.g., "Total 4 floors (-1,1,2,3)" indicates 1 basement floor and 1st to 3rd floors above ground).

Switch

Click the [Switch] button to set the map as the robot's current operating map, and the robot will perform handling tasks in the area corresponding to this map.

Edit

Click the [Edit] button to enter the map editing interface, where parameters such as the map's name and number of floors can be modified.

Delete

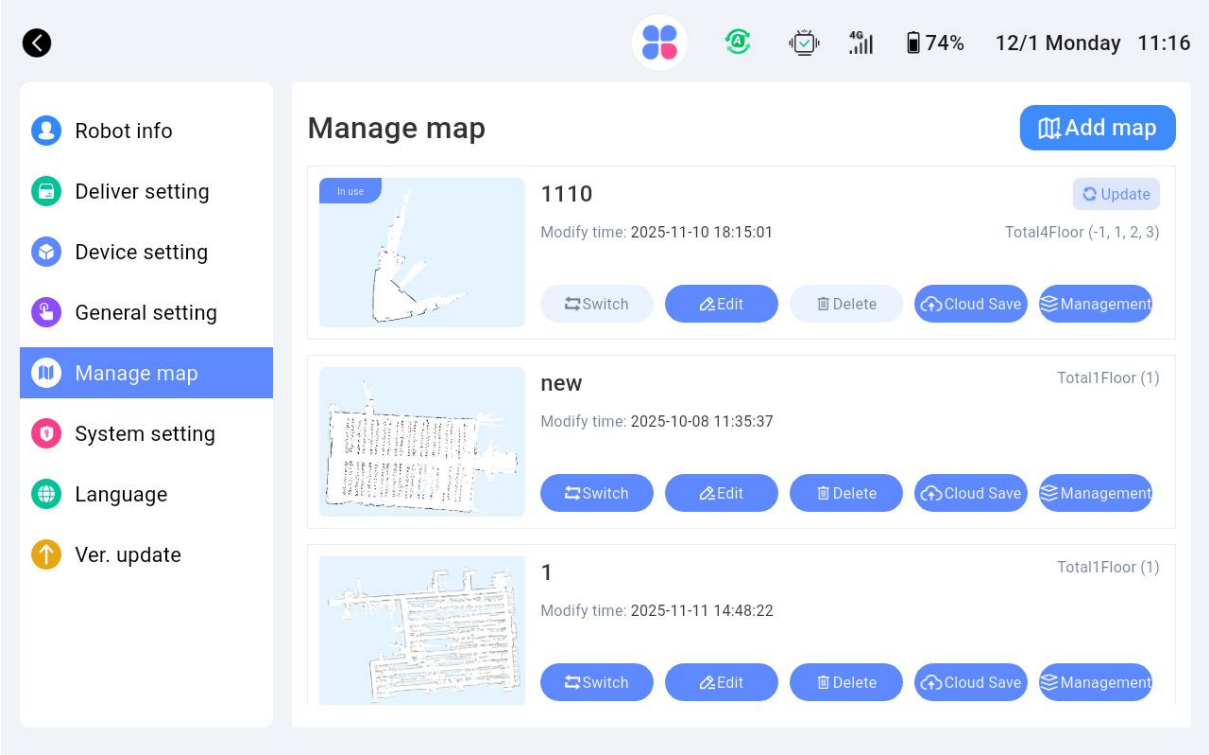
Click the [Delete] button to delete the map (Note: Deleted maps cannot be recovered; please operate with caution).

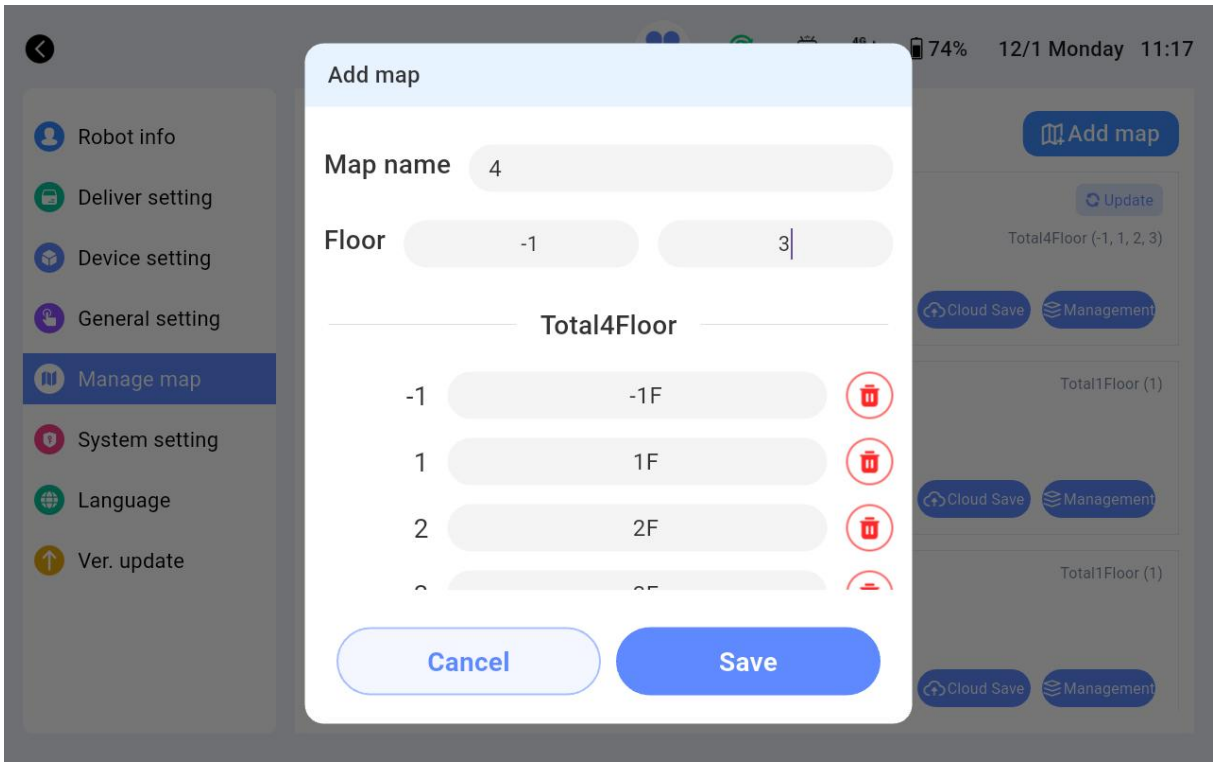
Upload to cloud

Click the [Upload to Cloud] button to upload the map to the cloud, enabling map sharing across multiple devices or cloud backup.

Floor management

Click the [Floor Management] button to manage the floor maps of the overall map.





Floor Map Management

As a sub-function of the "Map Management" module, Floor Map Management is designed for the refined management of single-floor operation maps for industrial handling robots. It supports operations including adding new floor maps, switching maps, viewing maps, editing maps, expanding maps, and deleting maps, ensuring the accuracy and operability of each floor map when robots operate in multi-floor scenarios.

Add new floor map

Click [Add New Floor Map] to create a new floor map, meeting the map expansion needs in multi-floor operation scenarios (e.g., adding floor maps for "1F", "2F", etc.).

Map thumbnail

Intuitively displays the layout of the floor map for quick identification.

Floor identifier

e.g., "-1F", which specifies the floor corresponding to the map ("-1F" denotes the 1st basement floor).

Modification time

e.g., "2025-11-03 09:54:21", records the time of the map's last modification to facilitate traceability of update records.

Switch

Click the [Switch] button to set the floor map as the robot's current operating floor map, and the robot will perform handling tasks on the corresponding floor according to this map.

View

Click the [View] button to check detailed information such as the layout and paths of the floor map, making it easy to confirm whether the map configuration meets operational requirements.

Edit

Click the [Edit] button to enter the map editing interface, where parameters such as the layout and paths of the floor map can be modified to adapt to changes in the operation area.

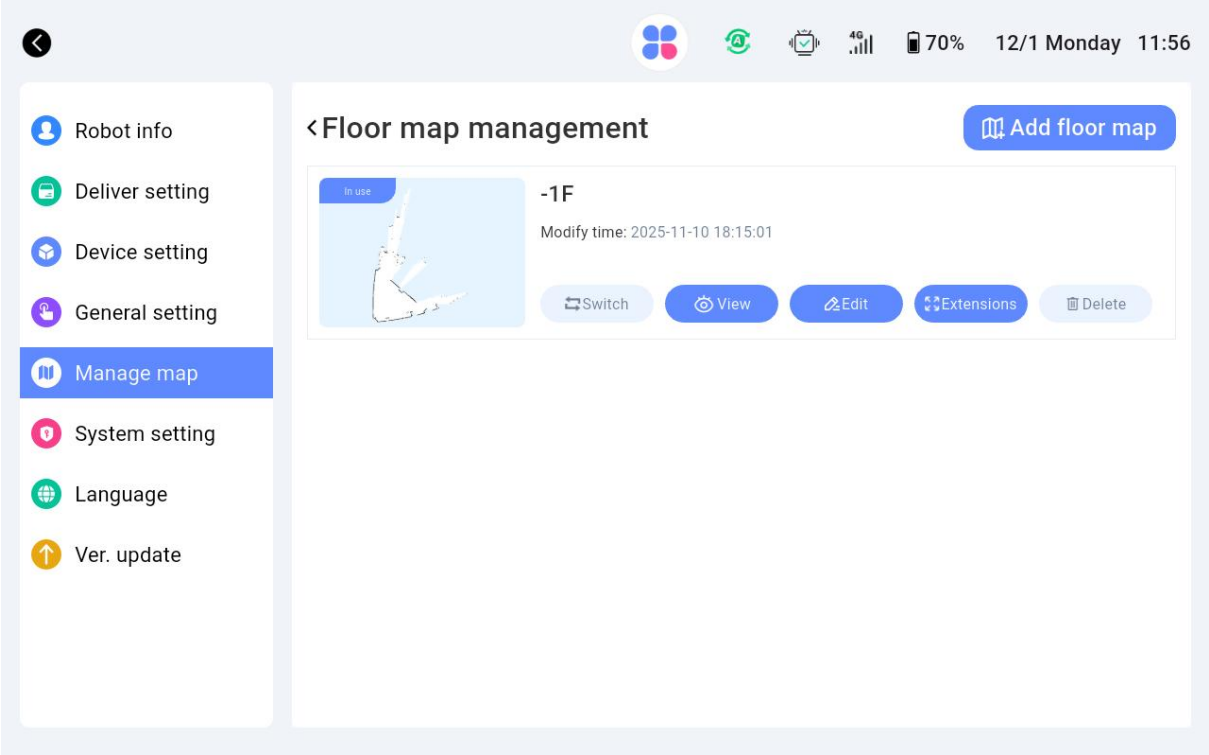
Expand

Click the [Expand] button to enable the robot's map expansion mode.

Delete

Click the [Delete] button to delete the floor map (Note: Deleted floor maps cannot be recovered; please confirm there is no further need for use before performing this operation).

The password for entering the settings can be changed.



Create new floor map

In multi-floor scenarios, you need to select the floor to be created and click [Start Creating New Map].

Mapping process

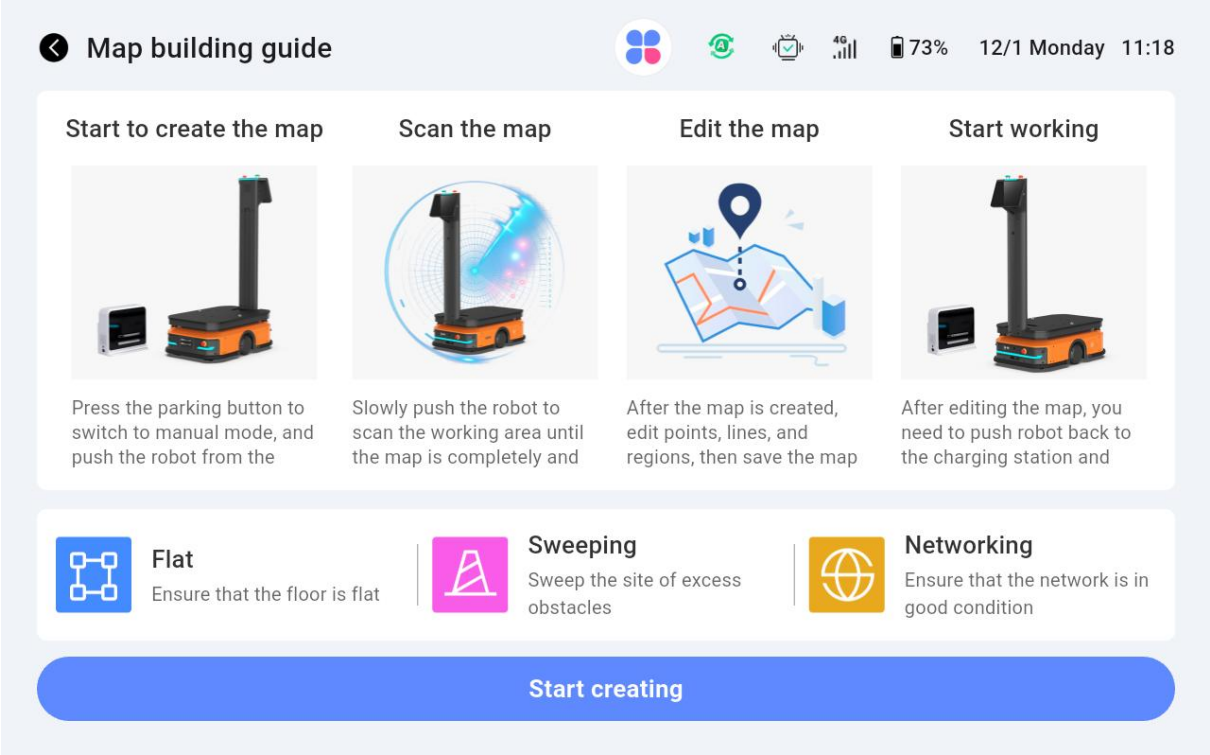
Manually push the robot slowly to cover all the routes it needs to travel. The system collects environmental data in real time, and the mapping trajectory is displayed synchronously on the interface.

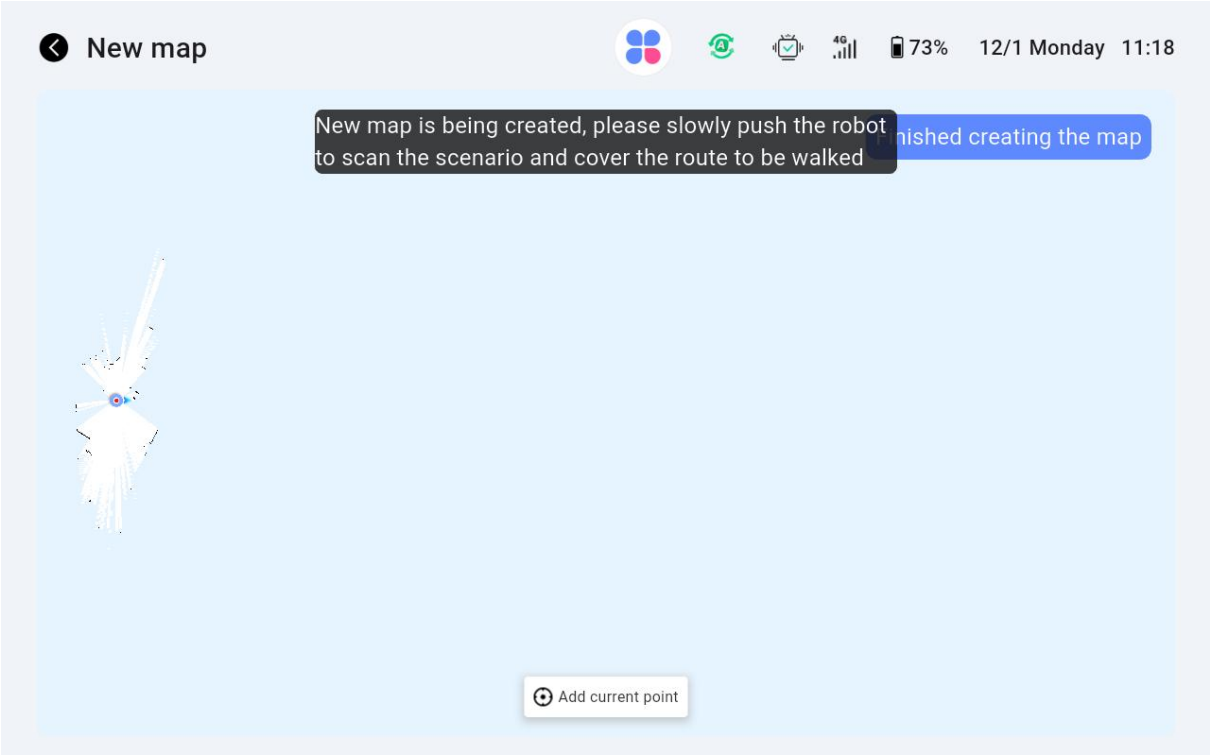
Add current point

During the mapping process, you can mark the current position according to the robot's pushed location.

End operation

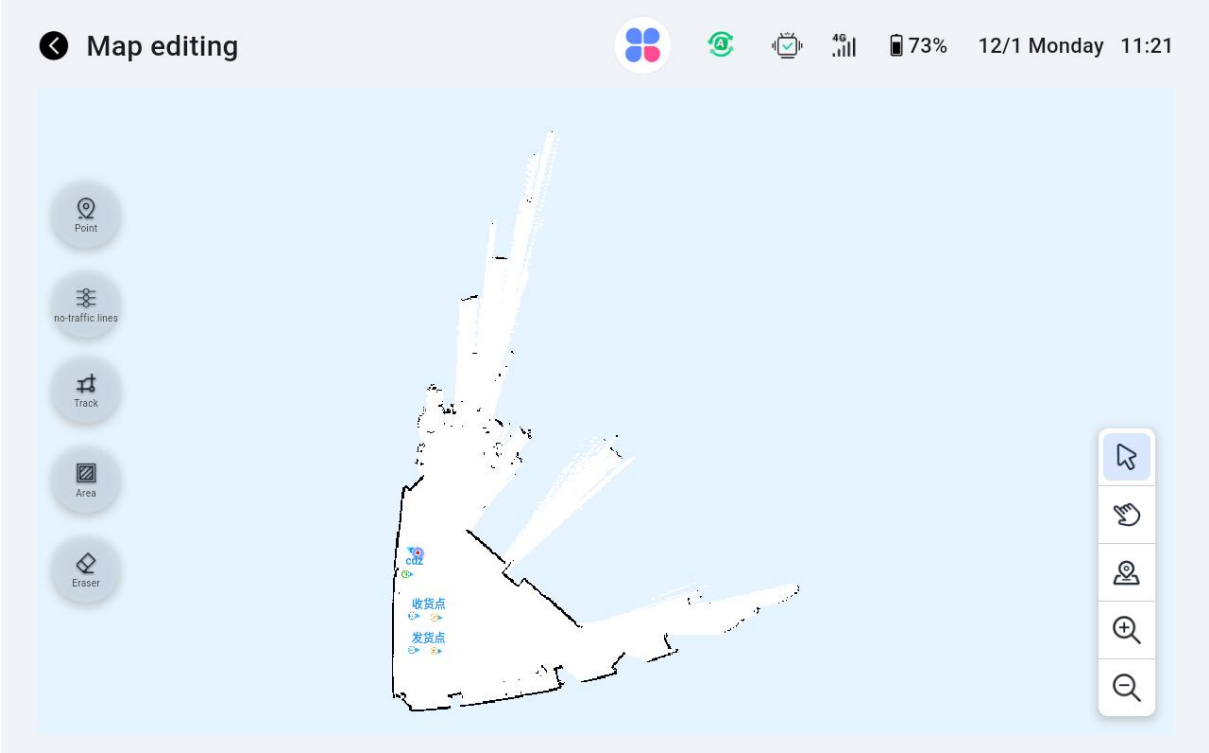
Once all routes are covered, click the [Complete Mapping] button to save the newly created map.



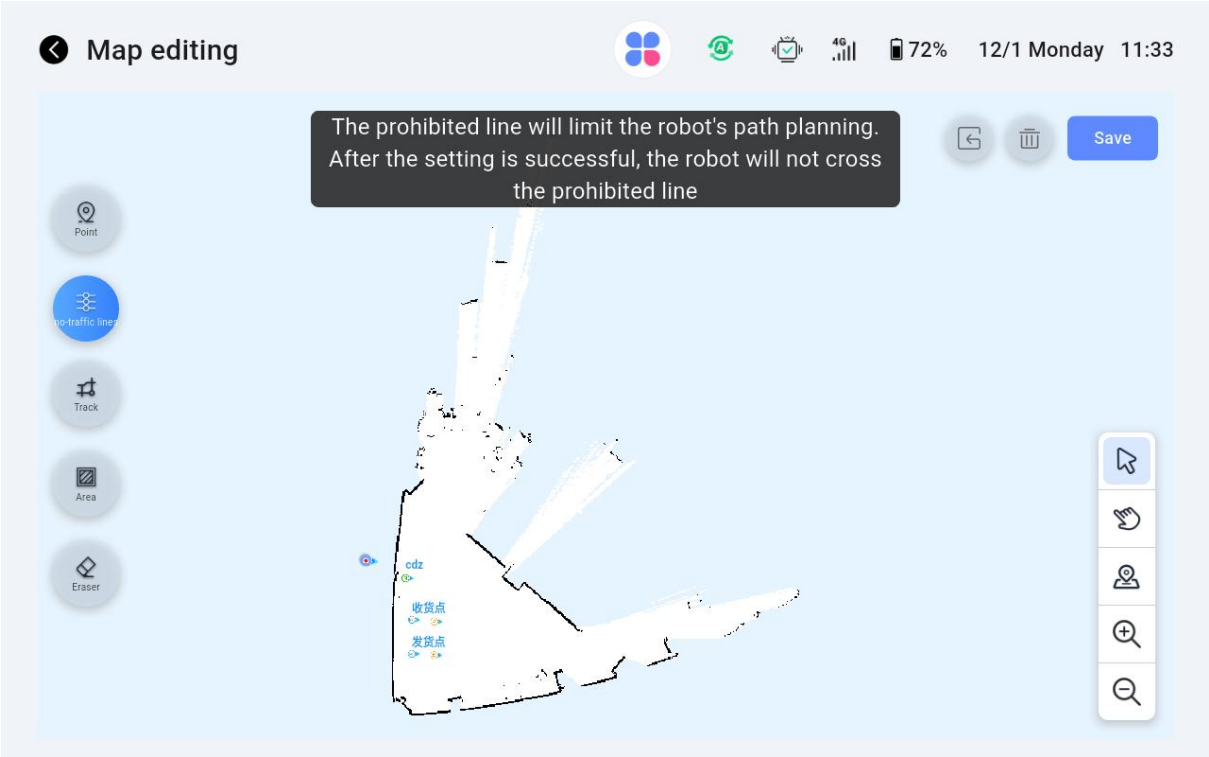


Left tools:

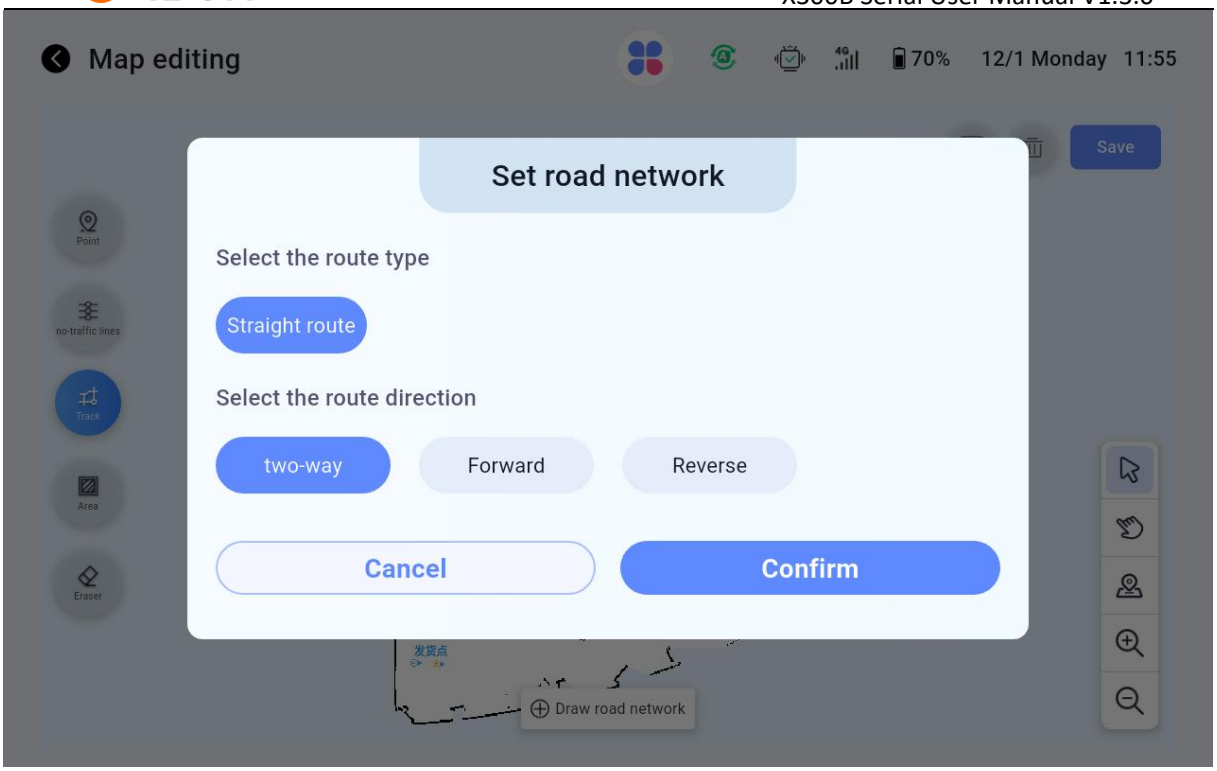
Point Position: Add the robot's task point position through the current point or a specified point, and set the point Angle.



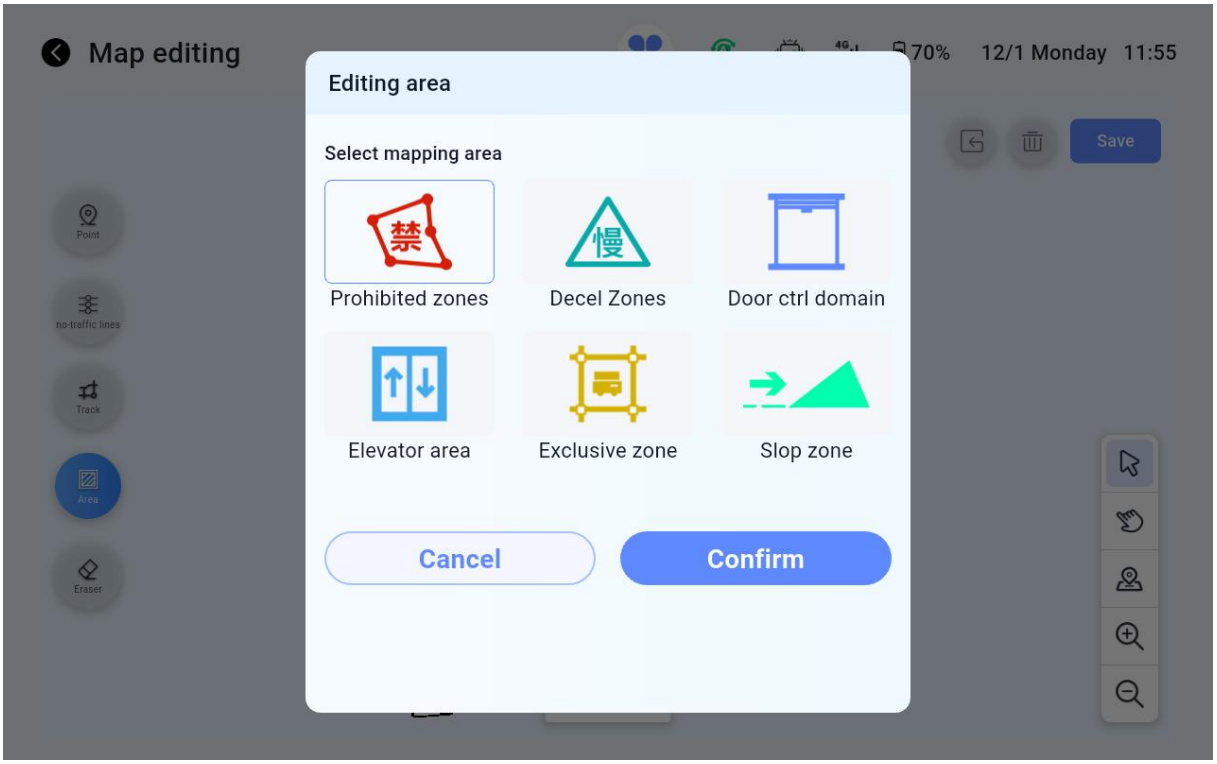
No-entry line: Draw the routes that robots are prohibited from passing through.



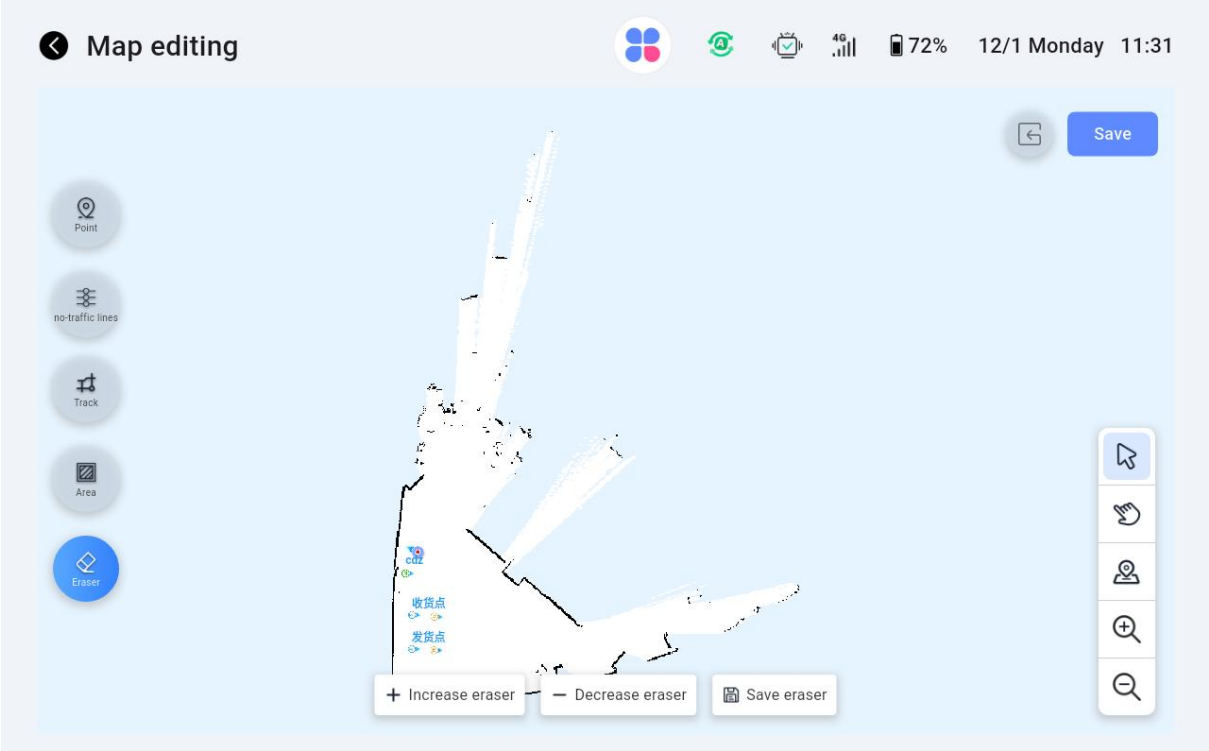
Road network: It can perform road network drawing operations. The default road network type is a linear road network, and the road network direction can be selected (bidirectional, forward, reverse).



Area: Multi-functional areas can be created (such as no-entry zones, deceleration zones, door control zones, elevator zones, exclusive zones, etc.).



Eraser: Remove obstacle from the map and optimize it.



Right side interaction: Supports editing selection, panning the map, locating the current robot position, and zooming in/out of the view, facilitating detailed editing

3.8.5 System Settings

Network settings

Wi-Fi connection configuration is supported.

System settings

Password Change: Allows modification of the access password for entering the settings.

Developer Mode: Provided for professionals to debug advanced configurations. Ordinary users should avoid arbitrary modifications to prevent system abnormalities.

Task Permission Settings: When enabled, a password is required to edit or delete entries in the task list.

Volume adjustment

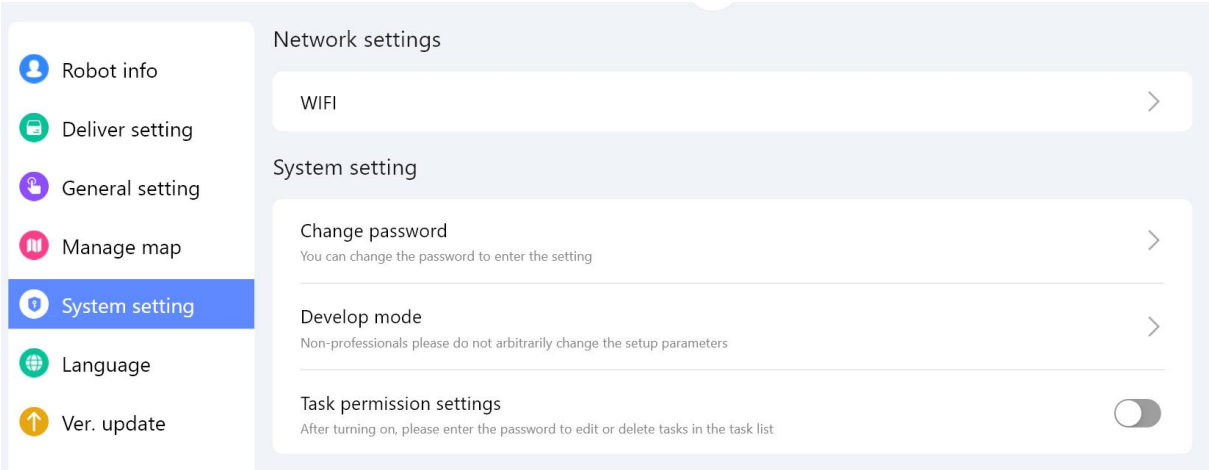
Adjust the system volume using a slider to flexibly control the device's sound level.

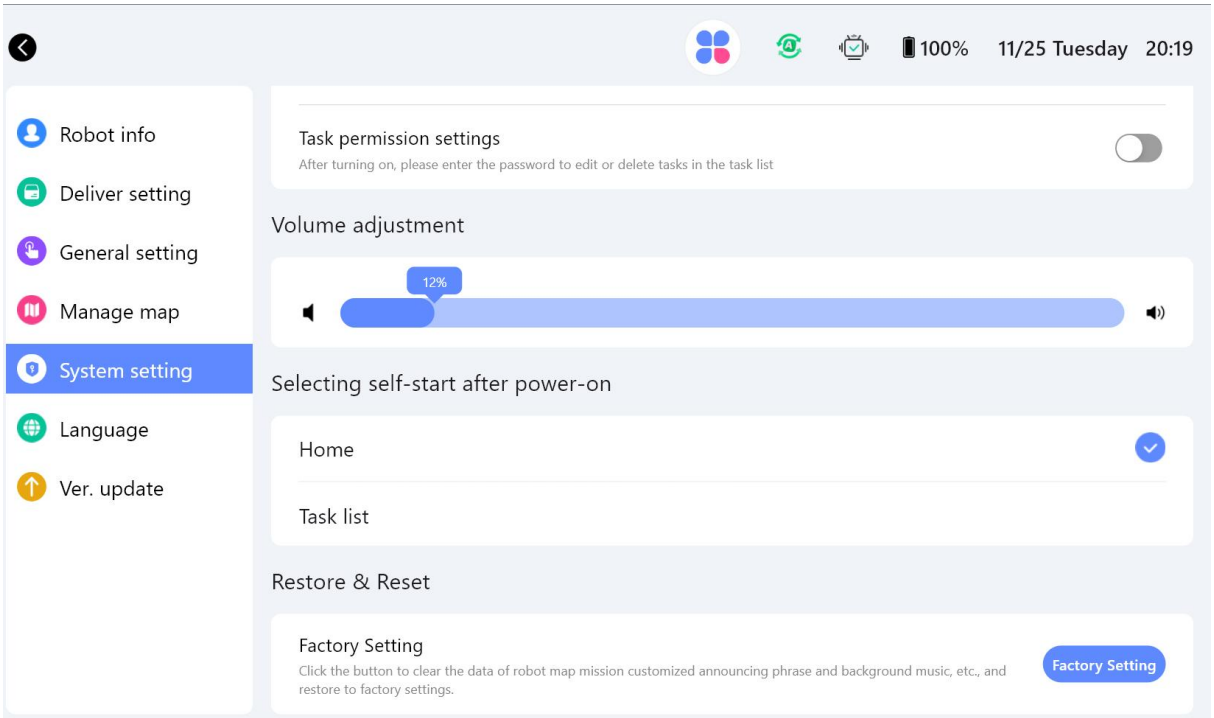
Select the seft-start after power

The seft-start module at boot defaults to the home page and can be switched to the task list.

Restore and reset

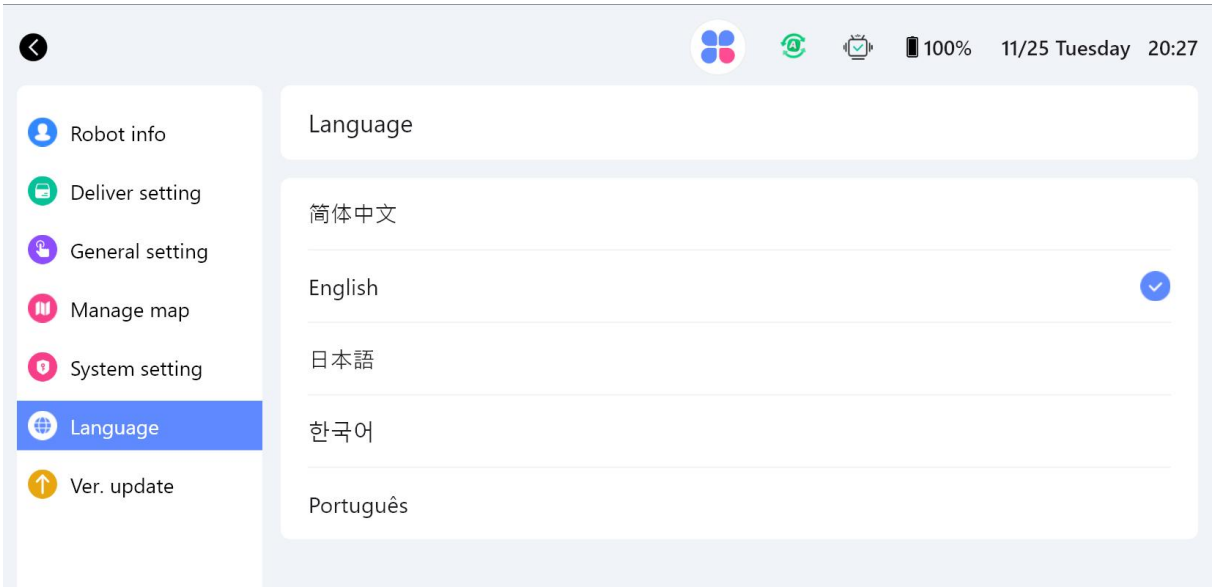
You can perform factory reset, clear the map, tasks, custom announcements, background music and other data with one click, restore the device to its initial state and restart it.





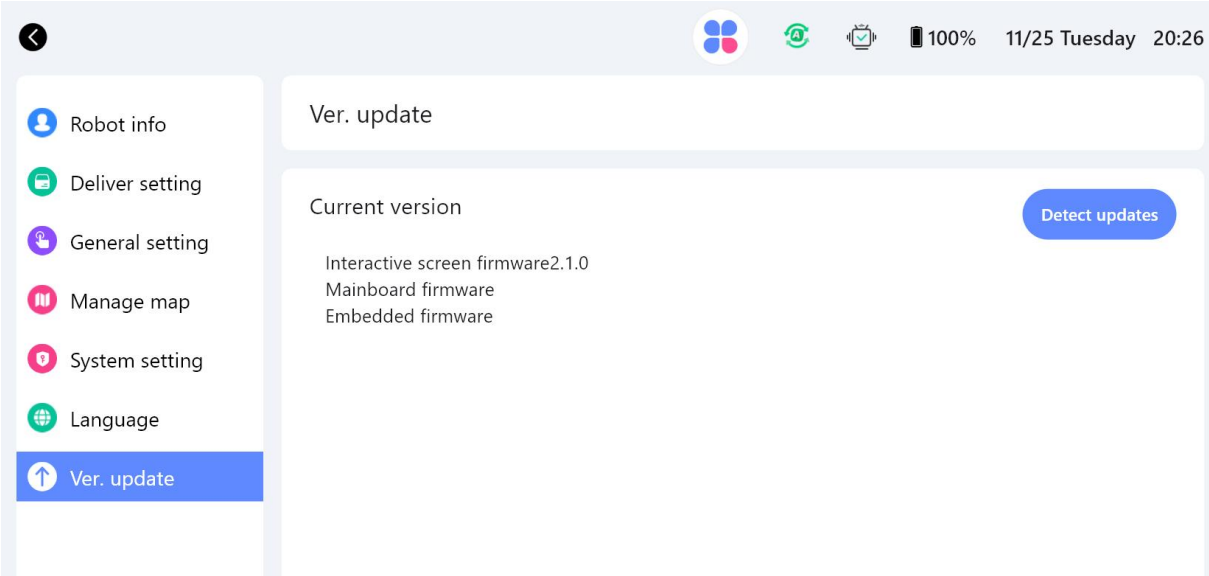
3.8.6 Language

Users can select the display language of the robot, providing language options such as Chinese, English, Japanese, Korean, portugues, etc. Interface languages can be switched as needed to meet the requirements of different language usage habits.



3.8.7 Version update

You can view the version information of the current interactive screen firmware, motherboard firmware and embedded firmware of the device. Click the "Check for Updates" button to check if there are any new versions, which is convenient for timely system upgrades.



4. Maintenance and common troubleshooting

4.1 Safety use

Please read this chapter carefully to fully understand the safety maintenance and troubleshooting process.

Only authorized personnel who have passed safety and other related training can maintain the robot system. Other related training includes robot system training and maintenance training conducted by manufacturers, distributors, and local importers.

Operators should participate in safety training in accordance with the regulations of each country.

	• Maintenance strictly in accordance with this manual. Do not remove

WARNING	or change any parts without authorization. Incorrect removal, change of parts or incorrect maintenance may cause abnormal operation of the robot and serious safety problems. ● When requires entering the robot work area in an emergency, please stop the robot. Otherwise, it may cause major safety problems. ● Please maintain and repair activities in the designated area. Before maintenance, make sure to turn off the robot to avoid sudden movement of the robot due to the scheduled task instructions. ● When it is necessary to maintain and repair the bottom of the robot, use equipment to lift the robot and ensure stability to avoid injuries caused by sudden falling of the robot.
 BE CAREFUL	● Do not put your hands between the pallet and the shell or between the pallet and the shelf. The relative movement of the pallet relative to the shell and the shelf may cause squeezing injury. ● If the robot switch is accidentally touched, the rotating parts such as the internal gears of the robot may cause squeezing or entanglement injury, and contact with the internal circuits may also cause electric shock. Therefore, when it is necessary to remove the shell, the please disconnect battery immediately to avoid the risk of injury, or notify the after-sales technician to operate.

4.2 Regular inspection and maintenance by the user

Recommended inspection of floor

Inspection items	Inspection period	Inspection method	Inspection content	Maintenance and repair methods

Channel area	Once a week	Visual	1. There is no extra matter or dust on the floor. 2. The floor is intact and not damaged.	1. Use a vacuum cleaner to remove extra matter and dust. 2. Use floor glue or cement to fill the damaged area.
Shelf area	Once a week			

Recommended inspection of shelf

Inspection items	**Inspection period**	**Inspection method**	**Inspection content**	**Maintenance and repair methods**
Shelf quality	Once a month	Visual	Whether the shelf is deformed or loose.	Repair and replacement.

Recommended inspection frequency of robot

Inspection items	**Inspection period**	**Inspection method**	**Inspection content**	**Maintenance and repair methods**
Robot power	Once a week	Check the robotic screen	1. Check the battery power on the screen.	1. Charge the robot by command or by pushing it to the charging station.
Shell	Once a month	Visual	1. The shell is free of dust and obvious dirt. 2. The shell is intact and not damaged.	1. Clean with a wet cloth that does not drip water. 2. If the shell is damaged, contact after-sales technicians to replace.

Inspection items	Inspection period	Inspection method	Inspection content	Maintenance and repair methods
LED light strip	Once a month	Visual	1.The strip lights up normally when the robot is turned on. 2. The light strip has the same color all around.	1. Check if the light strip connector is loose /replace the light strip that doesn't light up. 2. Replace the light strip with inconsistent colors.
Chassis bottom	Once a month	Visual	1. Check whether there is dust or extra matter on the chassis.	1. Use a vacuum cleaner to clean.
Driving wheel	Once a month	Visual	1. The wheels are free of extra matter. 2. The wheels are in good condition, with no damage or obvious wear.	1. Clean extra matter. 2. If the wheel is damaged or severely worn, contact after-sales technicians to replace.
Driven wheel	Once a month	Visual	1. The wheels are free of extra matter. 2. The wheels are in good condition, with no damage or obvious abrasion.	1. Clean extra matter 2. Replace the damaged or severely battered driven wheel.
LiDAR	Once a month	Visual	1. The lens is clean, free of dust or extra matter.	1. Clean with a soft cloth. If necessary, wipe with a non-woven cloth with anhydrous alcohol.
Charging pad	Once a month	Visual	1. The charging pad and electrodes are clean.	1. Clean with a clean soft cloth.

4.3 Instructions for use environment

- Do not use the robot or charge the robot in dangerous scenes such as high temperature, high pressure, or flammable and explosive environments to avoid personal injury or damage to the device.
- Do not use the robot in a humid, liquid or sticky environment to avoid damage to the robot.
- Do not use the robot in places where wireless devices are explicitly prohibited, otherwise it will interfere with other electronic devices or cause other dangers.
- Do not expose the battery to high temperatures or around heating devices, such as sunlight, heaters, microwave ovens, ovens or water heaters.
- This robot is suitable for use on relatively flat ground. It is not suitable for use on ground with high steps, large slopes and gaps in the environment.
- Do not dispose of the robot and its accessories as ordinary household waste. Please comply with local laws and regulations for the disposal of this robot and its accessories, and support recycling actions.
- Please use the robot on a flat, non-pothole-free ground. If the robot is stuck in a pothole, please push the robot chassis to get it out of trouble. Do not push or pull the robot hard.

5. Thank you for using our robots

Sincerely thank you for choosing our products! iBEN ROBOT will continue to iterate and upgrade its functions. If you have any questions or suggestions during use, please share them with iBEN ROBOT technical support staff or dealers. iBEN ROBOT will provide feedback as soon as possible. Your trust and support are our greatest motivation for progress. In the future, we will continue to polish our products with ingenuity and give you better services to meet your expectations. If you have any suggestions or needs, please feel free to contact us at any time - your satisfaction is always the direction of our unremitting efforts. I wish you a happy life and smooth work!